

TEDP-Rev C.



CRISTAR

University of Toronto

**Submitted in conformity with the requirements
for the CAN-RGX IX program**

Submitted: 08/12/2026

Executive Summary

The final TEDP for the CRISTAR project outlines our aim to study laser-induced crystallization under microgravity conditions as part of the CAN-RGX IX program. Crystallization is a critical process across multiple technological fields, and recent advances in laser-induced crystallization have enabled rapid, highly controlled crystal formation on Earth; however, the role of gravity in these processes remains poorly understood. By comparing the morphology and structural characteristics of crystals formed using laser-induced crystallization in a microgravity environment to those formed using the same technique on Earth, this project aims to advance the development of space-based crystallization techniques.

The CRISTAR team is composed of seven undergraduate students, two graduate physics students, and a supporting faculty advisor, all from the University of Toronto. To manage the technical requirements and timeline, the project is organized into five phases: design development, ground testing, pre-flight testing, flight campaign, and post-flight review. The team has completed the pre-flight testing. The flight test phase will take place on the week of August 31st, 2026. Each phase includes clearly defined objectives, deliverables, and responsibilities, supported by a detailed work breakdown structure, timeline, and contingency protocols. Flight safety, laser safety, and experimental reliability are prioritized throughout the project lifecycle.

The TEDP outlines the final technical design with CAD models, images of the final design, the results from the preflight testing which verified that the final project meets the project design requirements.

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1.0 Flight Manifest

First and Last Name	Level of study	Program	Email	Campaign Status
Gabriel Caribe	Undergraduate	Engineering Sciences	G.Caribe@mail.utoronto.ca	Mission Specialist (Primary)
Syeda Mahdia	Undergraduate	Engineering Sciences	Syeda.Mahdia@mail.utoronto.ca	Mission Specialist (Primary)
Elias Barsa	Undergraduate	Engineering Sciences	Elias.Barsa@mail.utoronto.ca	Ground Crew (Back up Mission Specialist)
Alexander Wainwright	Graduate	Physics	Alexander.Wainwright@mail.utoronto.ca	Ground Crew (Back up Mission Specialist)
Natalie Djuric	Undergraduate	Astro Physics	N.Djuric@mail.utoronto.ca	Ground Crew
Khaled Madhoun	Graduate	Physics	Khaled.Madhoun@mail.utoronto.ca	Ground Crew
Ian Mo	Undergraduate	Computer Science	Ian.Mo@mail.utoronto.ca	Ground Crew
Lauren Altomare	Undergraduate	Engineering Sciences	Lauren.Altomare@mail.utoronto.ca	Not Attending
Daniel Yu	Undergraduate	Computer Engineering	Danielz.Yu@mail.utoronto.ca	Not Attending

Note: The primary mission specialists will be on the first flight and back up mission specialists on the second flight.

2.0 Experimental Background

Crystallization is a foundational process in materials science, with applications in pharmaceutical drug development¹⁻⁵, organic electronics⁶⁻⁸, energy storage^{9,10}, and semiconductor manufacturing¹¹. Interest in space-based crystallization processes is on the rise and has developed into a distinct part of the space industry¹²⁻¹⁶. A promising new ground-based technique involves laser-induced crystallization¹⁷, where short-lived bubbles rapidly reduce pressure and trigger nucleation. This method shortens crystal formation time from hours to seconds, with enhanced control over crystal size¹⁸, offering potential for high-throughput production of nano- and microcrystals¹⁸. However, the role of gravity in these dynamics, especially in laser-induced methods, remains poorly understood. Clarifying this influence is important for advancing space-based manufacturing and improving Earth-based simulations of space-grown materials.

This research aims to demonstrate laser-based crystallization in a microgravity environment and to compare the resulting crystal structures with those formed on Earth. The experiment will assess whether laser-induced crystallization under microgravity conditions accelerates crystal formation and enhances sample quality relative to Earth-grown crystals. These insights are particularly relevant for producing crystals suitable for diffraction-based structural determination using electron beams, which are now sufficiently compact and portable for spaceflight applications¹⁹.

Key stakeholders include pharmaceutical companies that depend on high-quality crystals for drug development¹⁻⁵, Canadian startups looking for new niches in the space industry²⁰, and the Canadian Space Agency, which supports research that strengthens Canada's role in the emerging space economy. Environmental considerations are also important. The proposed laser-based crystallization approach requires only short suborbital flights to produce thousands of samples, rather than relying on traditional orbital missions. If successful, laser-induced crystallization techniques may provide an alternative pathway for some space-based crystallization applications that could reduce dependence on long-duration orbital missions.

The motivation behind this experiment is to lay the groundwork for scalable crystallization techniques in space, enabling space-based manufacturing, high-throughput screening and structural analysis using electron diffraction. By bridging fundamental science with engineering implementation, this research addresses critical questions about gravity's role in materials science.

The goal of the proposed experiment is to demonstrate the production of microcrystals using laser cavitation-based crystallization in a short-duration microgravity environment, and to compare them to crystals produced using the same technique under Earth's gravity. This objective includes demonstrating that laser-induced crystallization can reliably occur in microgravity and determining whether gravity influences the morphology and structural characteristics of the resulting crystals.

3.0 Experimental Design

To investigate the role of gravity in laser-based crystallization, the experiment uses a compact laser cavitation system designed for safe and reliable operation in a microgravity environment (see Figure 1). The system consists of five main subsystems: the laser optical system, the fluid handling system, cavitation monitoring sensors, the control and data acquisition system, and the enclosure.

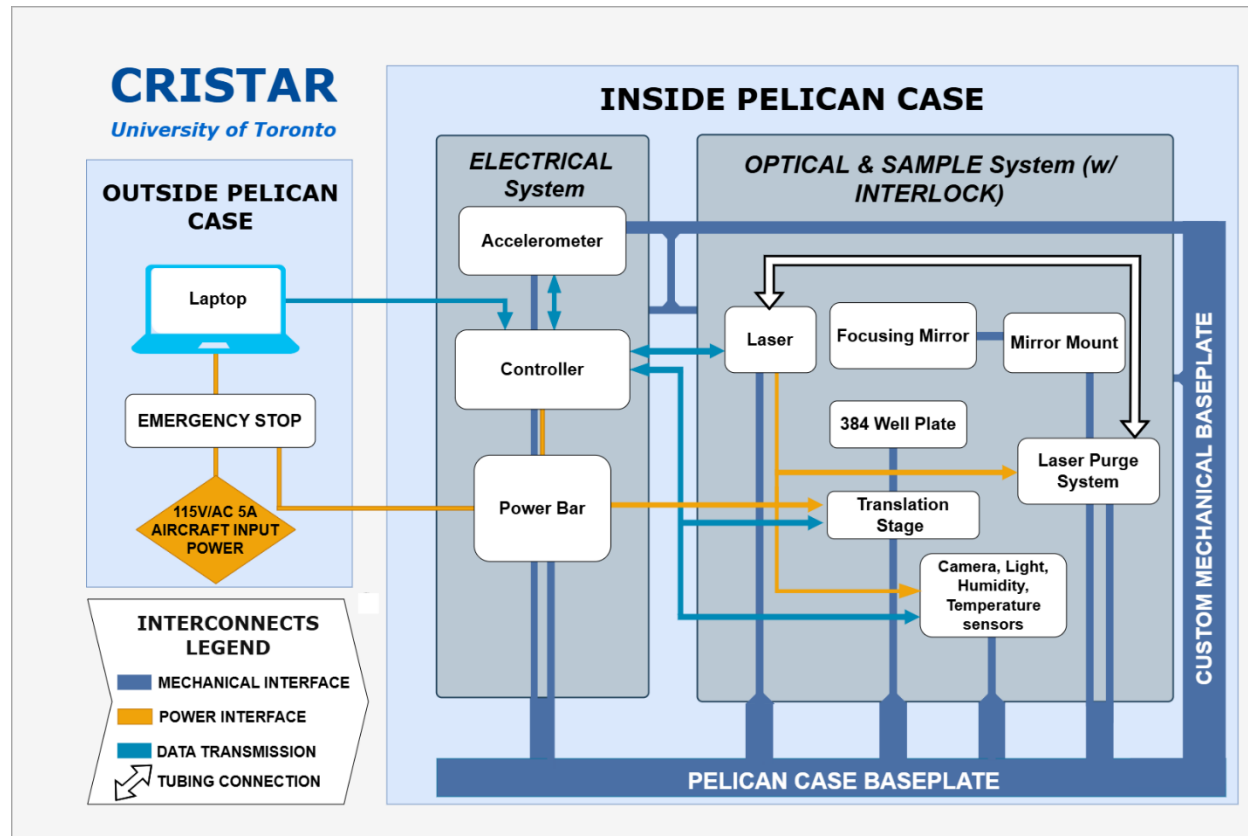


Figure 1: Block Diagram of the system architecture used to produce crystals using laser-induced crystallization.

3.1 Mechanical Mounting:

The laser optical system is rigidly mounted to the bottom panel of the Pelican case using vibration-damping brackets to minimize oscillations during parabolic maneuvers. The laser module is bolted directly to the base plate of the chamber, with a focusing mirror aligned along the optical axis. The focusing mirror is mounted to a holder which is made from PLA and attached directly to the base plate, ensuring stable alignment with the laser beam path. A 384-well plate, containing the crystallization solution, is positioned at the focus of the optical system on a motorized linear translation stage (one-dimensional translation with a minimum step size of 500 μm). The linear translation stage is secured to the pelican case base plate using threaded fasteners and vibration-isolating spacers. The laser beam is focused through an off axis parabolic mirror into the wells, with the optical path enclosed by a laser-safe housing with a magnetic interlock system. These interlocks ensure the laser can only operate when the housing is fully closed, maintaining Class 1 laser safety compliance.

To monitor the laser cavitation conditions an accelerometer, temperature and humidity sensor array is mounted to the supporting shelf inside the Pelican case. A CCD camera is used to monitor the

cavitation process and is fixed to the supporting shelf to provide a stable view of the well plate. An LED light is also included as part of the camera system and mounted to the supporting shelf to provide sufficient lighting inside the box.

3.2 Electrical and Data Connections:

Components draw power from the aircraft's 115 VAC, 5 A supply, routed through a power strip inside the case. The total system load is less than 600 W (see Table 3).

3.3 Laser Module:

The Ivy laser is powered from a single 24V output 90W power supply, equipped with a standard IEC320-C14 inlet and adapting to the input voltage in the range of 90VAC to 264VAC. The power supply contains an overload protection (hiccup mode), an over voltage protection (re-power required) and an over temperature protection (re-power required). Once the laser is powered, the internal electronics check that the interlock is closed. When they are open, laser emission is disabled. The laser will check all internal modules for proper communication and status, in this period the status LED is blinking in orange. Once all internal modules are cleared the laser is ready to receive commands and the status LED is breathing slowly in green. Note that the laser cannot be initialized until an external command is given and there is no manual way to initialize the laser head.

To initialize the laser, it must be put in "standby mode". In this mode the laser will thermalize, i.e. all components are brought to the defined temperature. During thermalization, the status LED is flashing green at a frequency of about 2 Hz and the GUI. This thermalization process can take up to 10 minutes, depending on the ambient conditions. Once thermalization is complete, the status LED will shine firmly green, and the laser is ready for emission. The laser is already rated laser class 3b in this mode.

The laser can be put into standby mode by either using the manufacturer's GUI or using a command passed through ethernet. For the CRISTAR project, commands are given from the Raspberry Pi located inside the pelican case. This Raspberry Pi is controlled through an ethernet cable which is connected to a laptop on the outside of pelican case. The connection is run through a VNC which give a virtual desktop like control of the Raspberry Pi. Without connecting to the raspberry pi, there is no control inputs given to the laser. As such, the laser cannot be put into "standby mode" until the command is given from the Pi, which passed to the Pi through the ethernet port throughput on the pelican case.

To meet program requirements, the laser is enclosed in a laser safe box with an interlock, meeting Class 1 laser compliance.

3.3.1 Laser interlock system:

The laser is equipped with an interlock connector. Unless pin 1 and pin 3 of the connector are connected, laser emission is disabled. This interlock connector should be connected to the user's safety circuit, in this case two magnetic interlocks (ENTRY-GUARD™ Magnetic Non-Contact Safety Interlock, ETG-INLK-L/R) on the inside of the pelican case.

Early experiments with the interlock system on the pelican case used magnets to bypass the interlock system for initialization. The laser was then shutdown and the bypass removed before closing the box to engage the interlock. Then the system was rebooted before the laser could identify that the interlock was broken. When interlock was left open for longer than about a minute

with the laser shutdown, the interlocks would need to be reset by cycling the key inside the pelican case, which required opening the interlocks. In the case of the microgravity flight campaign, the laser will have to be unpowered for an extended period before integration, meaning that an interlock bypass will be required to allow for the key at the back of the laser to be cycled before the box is closed.

The interlock bypass must ensure that the laser cannot be initialized or operated while the box is open. We accomplish this through the following system:

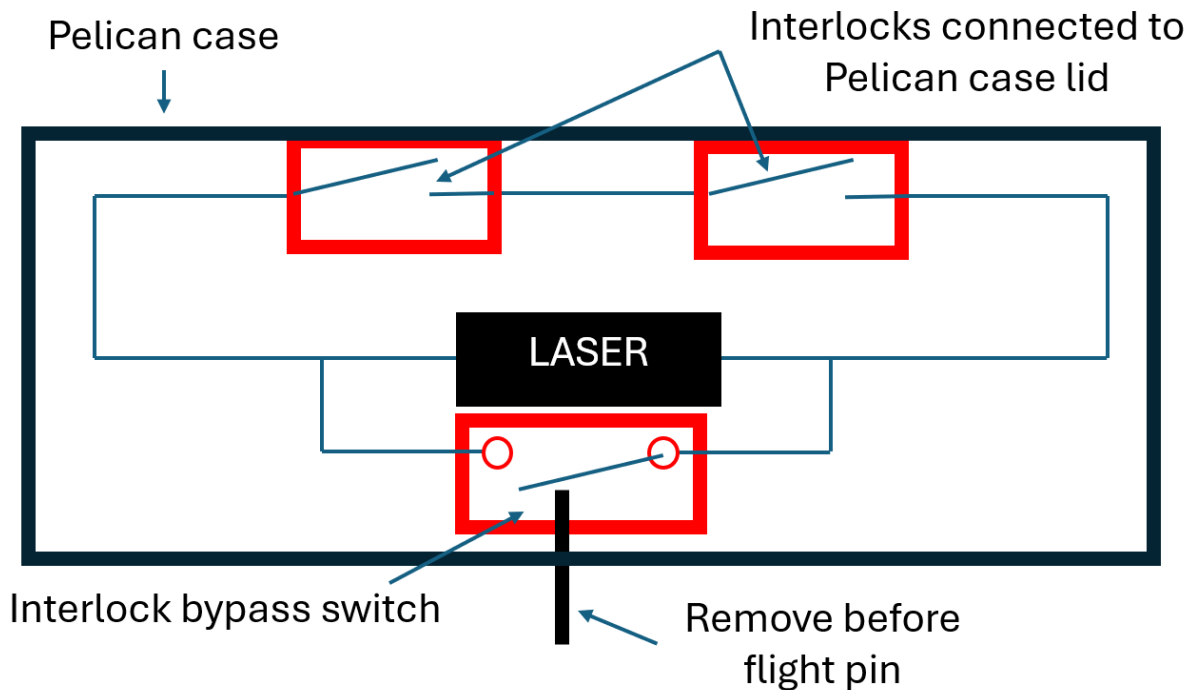


Figure 2: Conceptual diagram of the interlock system including a bypass for laser initialization.

Where the remove before flight pin blocks the ethernet connection port to the raspberry pi, preventing the laser from being enabled or thermalized when the interlock bypass is closed. The interlock bypass switch is configured such that the interlock bypass switch cannot be reset while the pelican case is closed. As such, it is not possible to open the box with the laser enabled or thermalized as opening the box triggers the interlock, shutting down the laser.

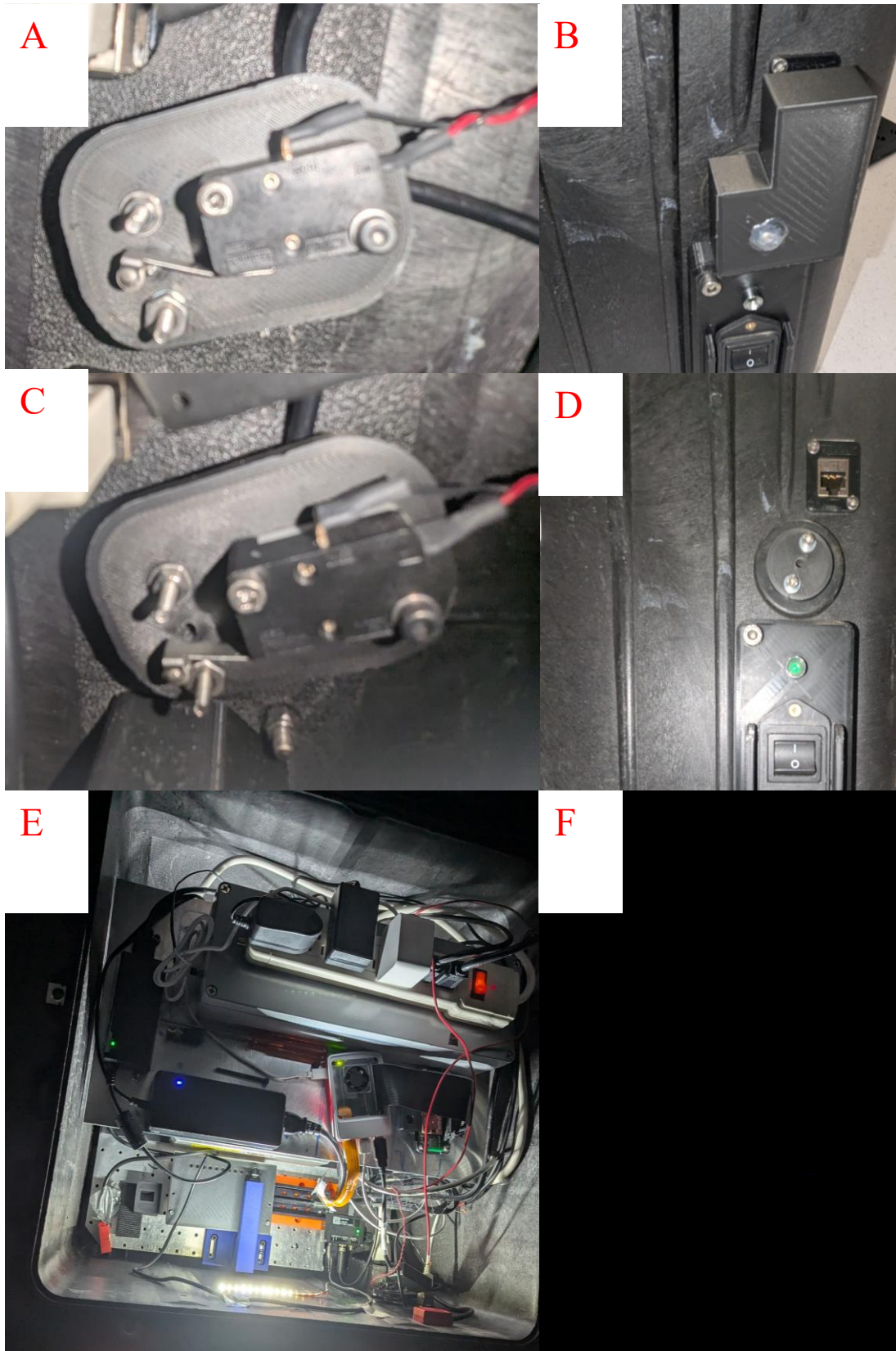


Figure 3: Photographs of the prototype laser bypass system. (A) Laser bypass activation switch with the external pin inserted, holding down the interlock button. (B) External pin installed through the Ethernet passthrough port, maintaining the interlock in the engaged position during laser startup. (C) Laser bypass activation switch with the pin removed, releasing the interlock button. The actuation lever is positioned below the pin hole, preventing re-engagement of the bypass without opening the enclosure. (D) External view of the passthrough port. (E) System during operation with indicator light on. (F) Open pin port in a dark environment, demonstrating that no light escapes through the port.

3.3.2 Laser purge system:

To ensure that the laser has a constant power output the relative humidity inside the laser head must be kept below 20%, per manufacturer guidelines. A home build dry air purger (purger) is used to control the humidity inside the laser head. The purger consists of a small air pump and an airtight cavity filled with silica gel bead size and an air filter to prevent dust entering the laser cavity. The silica beads are replaced once the laser head humidity exceeds 15% and has been demonstrated to maintain humidity below 20% for ambient humidities as high as 60%.

3.4 Crystallization solution:

Two standard biomolecules will be used in this experiment; hen egg-white lysozyme (HEWL) and Glycine. Both the HEWL and glycine solutions have been found to be shelf stable for a minimum period of 14 days. Two plates of each sample will be prepared with rows A and B filled. Row A will be used for laser irradiation while row B will be used as a nonirradiated standard of comparison.

3.4.1 HEWL Preparation:

HEWL was purchased from Sigma-Aldrich will be used without further purification (12650-88-3). Six stock solutions will be prepared by dissolving 100, 200, 400, 600, 800, and 1000 mg of HEWL in 10 mL of 0.1 M acetate buffer (pH 4.35), followed by filtration through a 0.22 μm syringe filter to remove particulates. Each solution will then be pipetted into the wells of a 384-well plate with a glass bottom (Corning® 384-well High Content Imaging Glass Bottom Microplate), completely filling each well. To prevent evaporation, a sterile tape is placed over the top of the plate. The well plate is then mounted inverted in the sample holder, with the glass face pointing towards the focusing mirror.

3.4.2 Glycine preparation:

Glycine was purchased from Sigma-Aldrich (56-40-6) will be used without further purification, and dissolved in heated distilled water at a concentration of 4.6 M. The heated solution was pipetted into the wells of a 384-well plate with a glass bottom. To prevent evaporation, a sterile tape is placed over the top of the plate. The well plate is then mounted inverted in the sample holder, with the glass face pointing towards the focusing mirror.

3.5 Control Interfaces:

The laser and translation stages are connected to an intermediate controller (Raspberry Pi) via USB connections. The intermediate controller communicates with the control laptop via Ethernet cable, enabling command execution and status feedback.

3.6 Sensor Data Interfaces:

The camera streams, translation stage, and sensor data is measured onto the Raspberry Pi computer. The Raspberry Pi is interfaced using the laptop onboard the aircraft with an ethernet cable through to the inside of the box. As seen in Figure 4, sensor data comes from two breakout boards (an inertial measurement unit and a barometer) assembled into a printed circuit board (PCB) measuring acceleration, velocity, orientation, temperature, pressure, and altitude. Both boards transmit real-time data to the Raspberry Pi via onboard input/output pins.

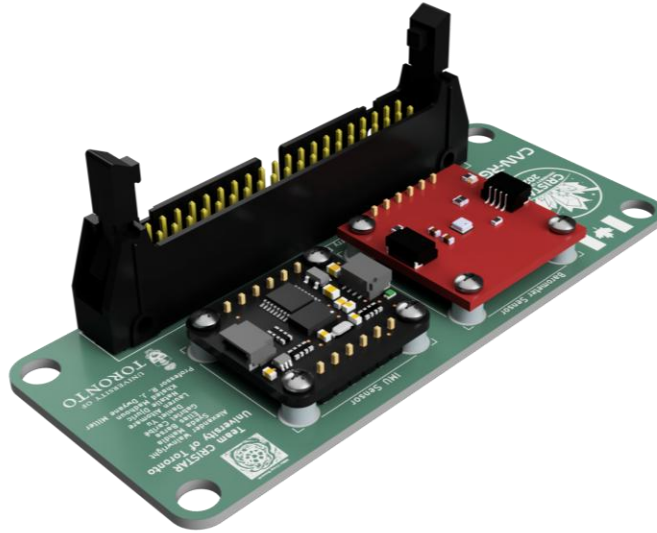


Figure 4: PCB model holding the two sensor boards.

3.7 Cable Management:

The Ethernet cable is routed through a bypass in the Pelican case with strain relief clamps to prevent disconnection during high-g maneuvers. Inside the case, wires are attached to the supporting shelf using Gorilla tape, zip ties, and screw mount cable clips to prevent disconnection and movement during flight operations.

3.8 Control and User Interface:

The mission specialist operates the system through a custom graphical user interface (GUI) on the control laptop. The GUI integrates all functions, including laser firing, stage positioning, and real-time monitoring of camera and accelerometer data. The laptop supervises the intermediate controller for laser and stage operations while directly logging sensor data. Data from the accelerometer and camera are time-synchronized and recorded for future reference with crystallization events.

4.0 Final Apparatus Design:

Final apparatus was constructed prior to preflight testing. To provide clear images of the design we provide information both on the CAD model and final physical design.

4.1 Computer Aided Design Model:

To facilitate design, a full CAD model of the integrated system has been developed. In summary, the system consists of two layers. The lower layer contains the optical and laser apparatus while the upper layer houses the supporting electronics. Figure 5, shows a cut view of the Computer Aided Design (CAD) model, with the position of the laser interlocks within the box shown in Figure 6. The focusing optical mirror, mirror mount and translation stage are shown in detail in Figure 7.

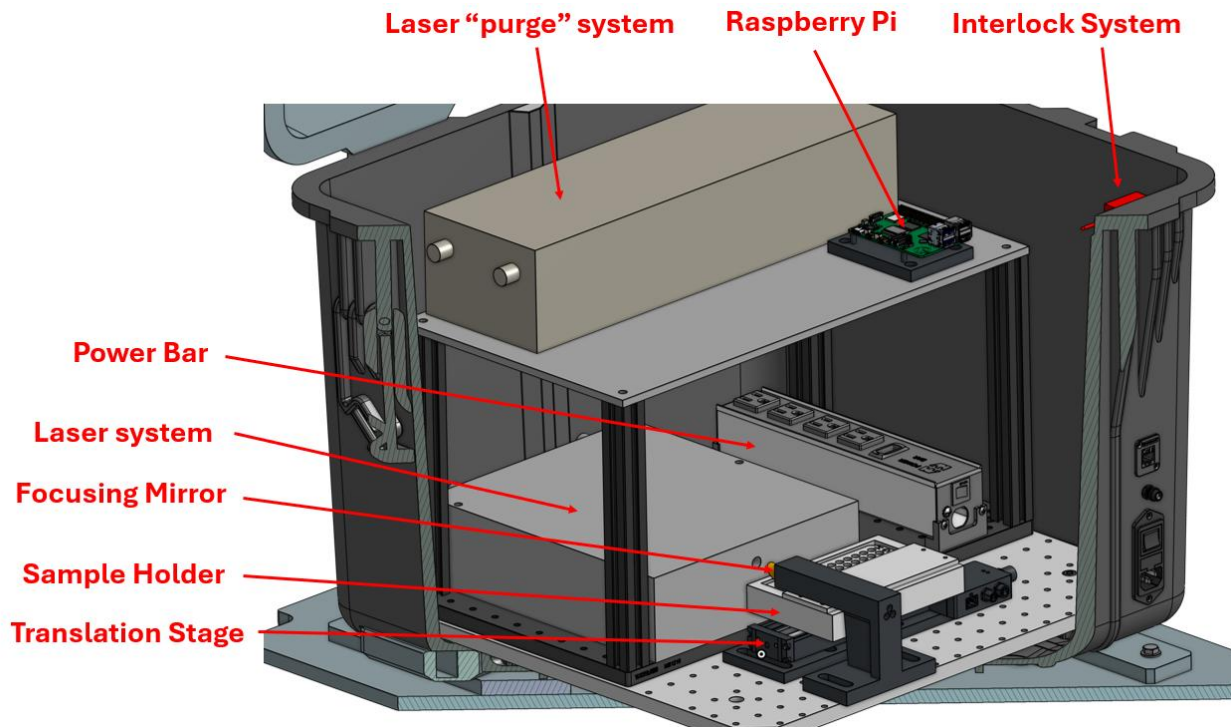


Figure 5: Side cut view of the CAD model of the pelican case structure showing the interlocks and laser system in blue.

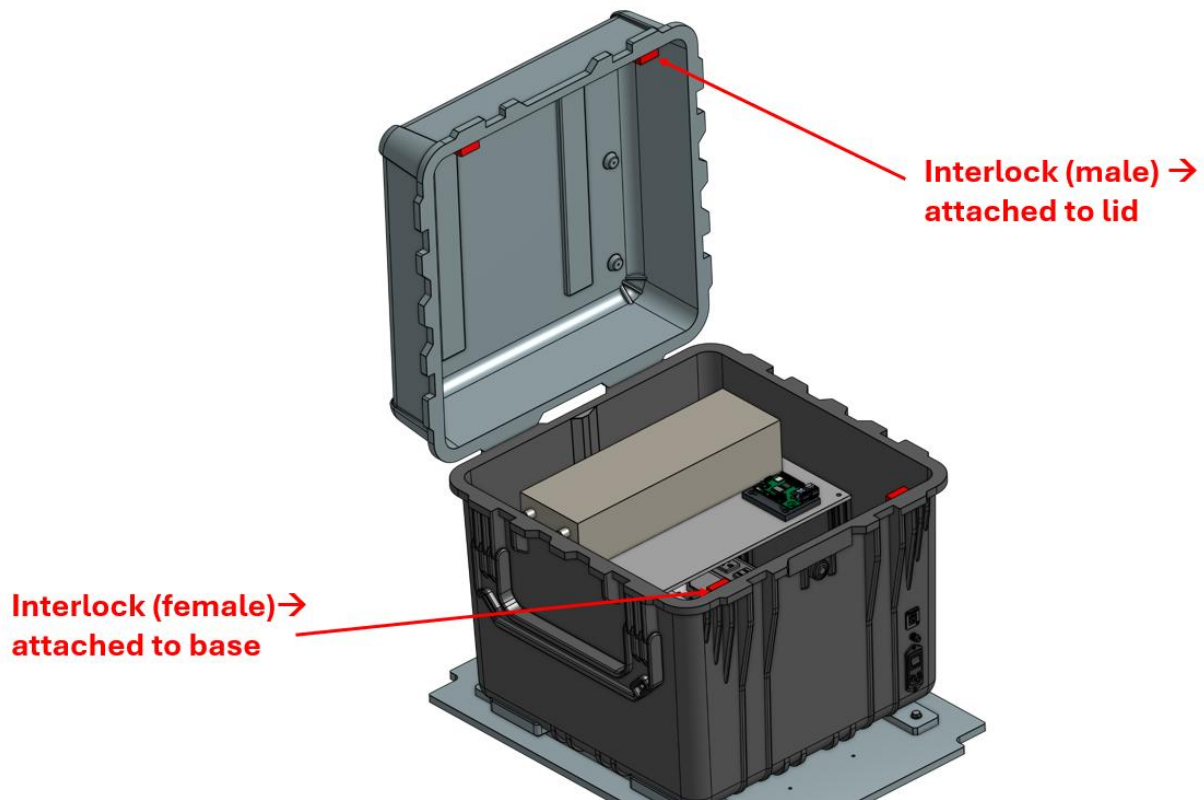


Figure 6: Interlock system for the pelican case enclosure.

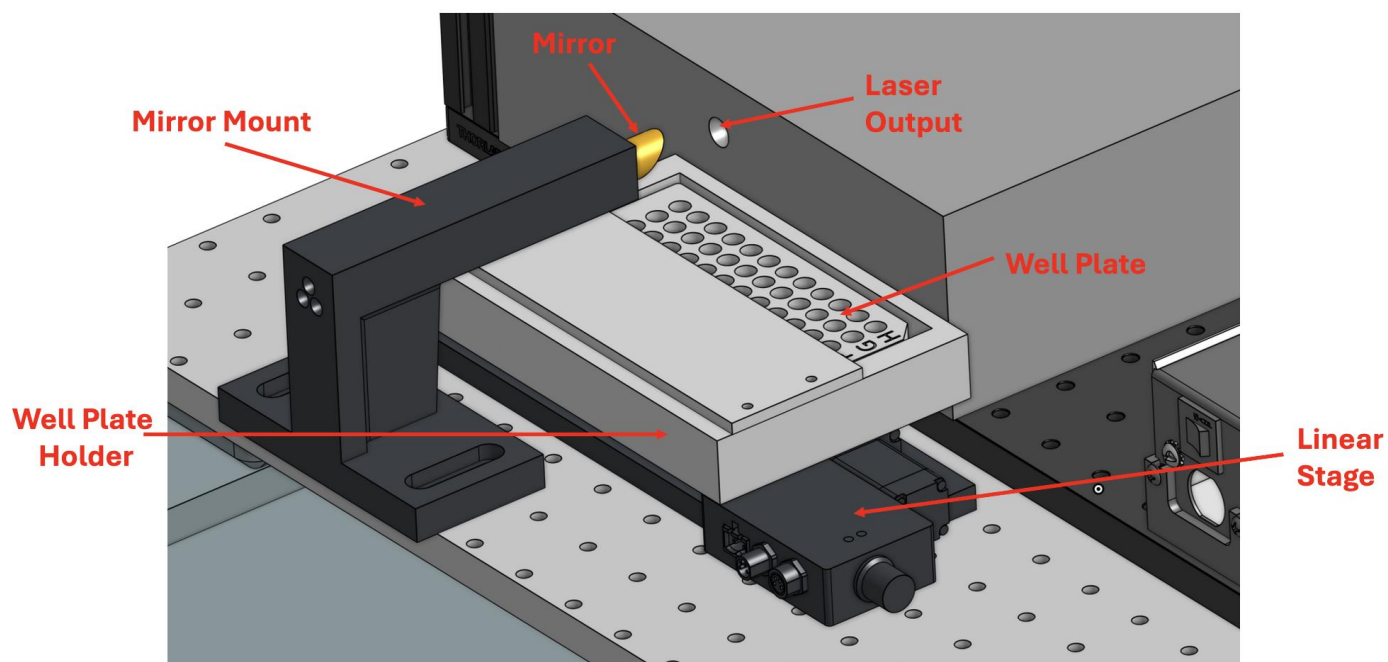


Figure 7: CAD model of the sample and translation stage system.

4.2 Final physical design:

The final physical system is shown in *Figure 8*, and closely resembles the CAD model above.

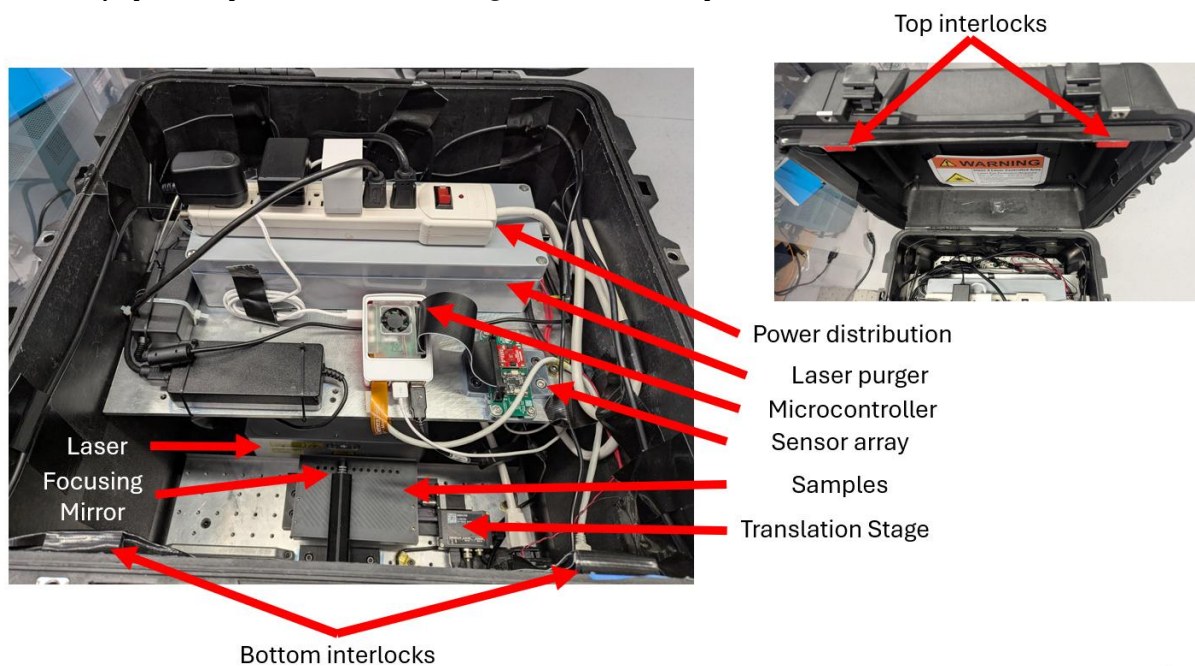


Figure 8: Final Physical design.

5.0 Bill of Materials:

Table 1 provides the list of materials to be brought onboard the NRC Falcon 20.

Table 1: Bill of materials for CRISTAR microgravity Flights.

Item Number	Sub System	Item	Description
1.	Optical	Laser	Glucoloop AG IVY Laser (2.1 ns, 2723 nm, 31 mW. 1.5 kHz)
2.		Mirror Mount	Custom PLA printed plastic mount.
3.		Off axis parabolic Mirror	1" EFL, Unprotected gold mirror (MPD019-M03)
4.		Magnetic Interlocks	ETG-INLK-L & ETG-INLK-R
5.		Purge system	Home built closed loop air drying system using silica beads and a low power air pump.
6.	Sample	384 well plate holder	Custom PLA printed plastic
7.		384 well plate	Corning® 384-well High Content Imaging Glass Bottom Microplate
8.		Protein Sample	25 mg/ml Egg White Lysozyme, 0.3-0.6 M NaCl, 0.05 M acetic acid. 80 µL per well.
9.		Small Biomolecule Sample	4.6 M Glycine in distilled water. 80 µL per well.
10.		Translation stage	1D Translation stage (Zaber, X-LSM100A)
11.	Control and monitoring	Raspberry Pi	Raspberry Pi 5
12.		Laptop	Lenovo IdeaPad Slim 3
13.		Camera holder	Custom PLA printed plastic
14.		Camera	Raspberry Pi Camera Module 3
15.		LED light	LED lighting strip
16.		Humidity and temperature sensor	SparkFun BME280
17.		Accelerometer	Adafruit BNO085
18.	Power distribution	Laser Power Supply	24 VDC, 3.5 A power supply
19.		Power adapter	See electrical load table
20.	Mis. Experimental Apparatus	Wires	Various gauges
21.		Fasteners	Various aluminum metric screws.
22.		Two-Sided Tape	Gorilla Heavy Duty Double Sided Mounting Tape
23.		Zip Ties	14-inch Cable Ties Indoor/Outdoor, Model XLS-4.8X370-0-C
24.		Gorilla Tape	Gorilla Utility Duct Tape
25.	Media and outreach	Cell Phone	Pixel 7, iPhone 13, iPhone 14 Pro Max iPhone 15 Pro Max

6.0 Mass Budget:

The mass budget, excluding the pelican case is provided in Table 2. With a total mass of 12.68 kg, less than 30% of the maximum 45 kg total weight is used.

Table 2: Mass Budget for CRISTAR Project.

System	Component	QTY	Unit Mass [kg]	Total Mass [kg]	Fraction of Total [%]	Verification Method
Structure and Mechanisms	Mirror Mount	1	0.2	0.2	2%	M0
	384 well plate holder	1	0.1	0.1	1%	M1
	384 well plate	1	0.1	0.1	1%	M1
	Camera holder	1	0.02	0.02	0%	M0
	Power Supply Holder	1	0.1	0.1	1%	M0
	Translation stage	1	0.4	0.4	3%	M1
Power Systems	Laser Power Supply	1	0.3	0.3	2%	M1
	9V Power Adapter	2	0.1	0.1	1%	M1
	Translation Stage Power Supply	1	0.3	0.3	2%	M1
	Raspberry Pi Power Supply	1	0.1	0.1	1%	M1
Command and Data Handling	Raspberry Pi	1	0.1	0.1	1%	M1
	Camera	1	0.01	0.01	0%	M1
	LEDs	1	0.05	0.05	0%	
Optical system	Laser	1	4.3	4.3	34%	M1
	Mirror	1	0.1	0.1	1%	M1
	Purge System	1	4.0	4.0	32%	M1
	Interlocks	2	0.05	0.1	1%	M1
Misc.	Sample	1	0.0	0.0	0%	M0
	Fasteners	18	0.1	1.8	14%	M1
	Wires	5	0.1	0.5	4%	E
			Total Mass:	12.68	100%	E
			Maximum Mass:	45	-	
			Mass Margin:	32.32	72%	

7.0 Electrical systems:

To enable the experiment, several different electrical systems must be integrated including the laser system, control system, and sensors. Details of the system are described below.

7.1 Schematic: A graphical representation of the electrical system is given in Figure 9. Power supply wires to electrical components are commercially supplied by the manufacturers of the translation stage, raspberry pi and Laser respectively. The laser system and translation stage power supplies have built in surge protection which should be sufficient to prevent damage to the component from fluctuations in the aircraft power. As all power supply and grounding is being done by the commercial manufacturer of commercial-off-the-shelf components used, no additional grounding is planned.

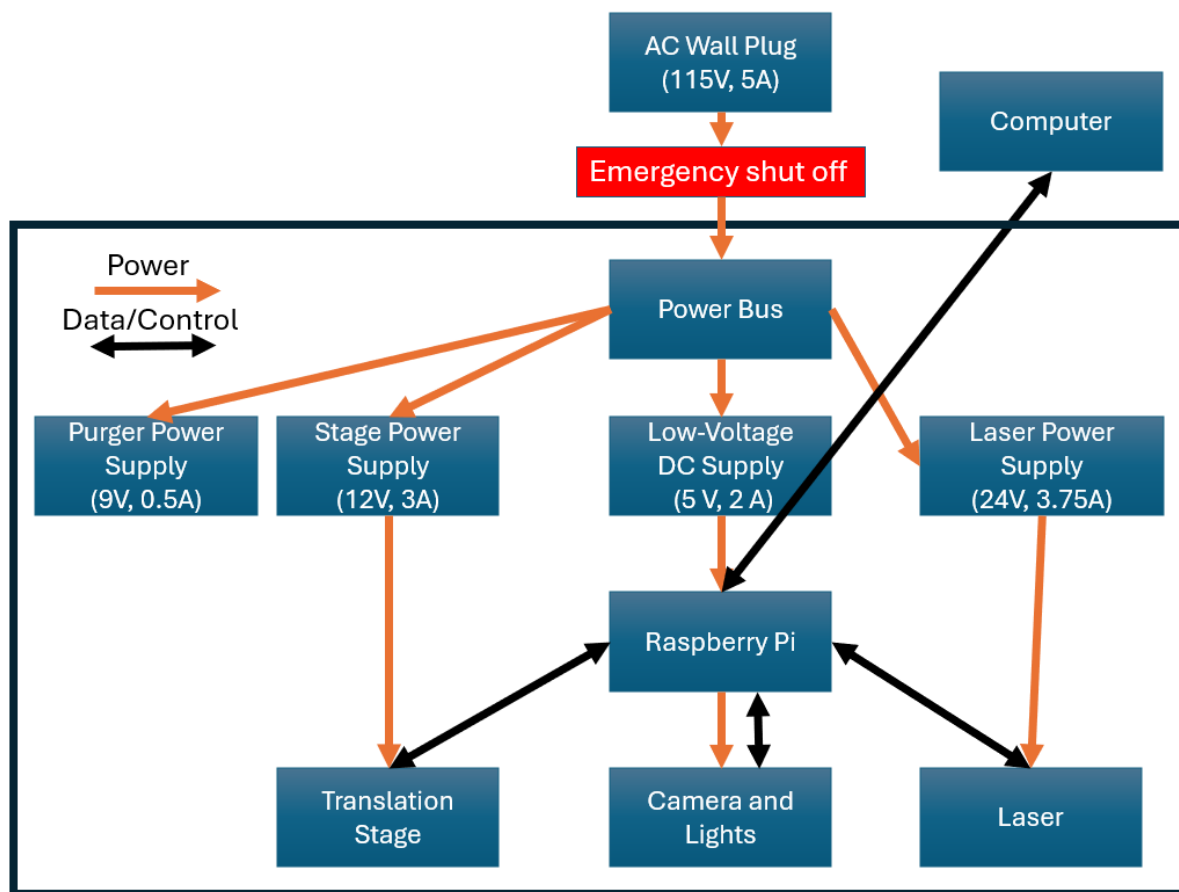


Figure 9: Graphical representation of the electrical power and control system featuring exclusively commercial off the shelf components.

7.2 Load Table:

The total electrical load is broken down by component in Table 3 with a total peak power consumption of 352.4 W, leaving 247.6 W of the available 600 W in reserve.

Table 3: Electrical load table for the CRISTAR project.

Component	QTY	Voltage [V]	Current [A]	Idle Operation		Science Operation		Verification Method
				Duty Cycle [%]	Power [W]	Duty Cycle [%]	Power[W]	
Laser	1	24 VDC	3.75	0	0.00	100	90	M1
Translation Stage	1	24 VDC	0.35	0	0	100	8.4	M1
Laptop power supply	1	120 VAC	2.00	100	230	100	230	M1
Raspberry Pi with sensors	1	5 VDC	3.00	100	15	100	15	M1
Laser “purge” system	1	9.0 VDC	0.50	100	4.5	100	4.5	M1
LEDs	1	9.0 VDC	0.50	100	4.5	100	4.5	M1
Total Power Consumed: [W]					254.00		352.4	
Total Available Power [W]					600.00		600.00	
Power Margin [W]					+346		+247.6	

7.3 Stored Energy:

No energy storage devices will be used in this RGX project.

8.0 Science Traceability Matrix Template

Table 4: Science traceability matrix for microgravity laser induced crystallization.

Scientific Objective	Scientific Measurement		Scientific Instrument	
	Measurement Objective	Measurement Requirement	Instrument	Instrument Requirement
Quantification of crystal size	Determine if gravity changes the crystal size and shape.	Average and maximum crystal size after formation in microgravity, hyper gravity, and normal gravitational conditions.	Electron microscope	Electron microscope resolution below 1 nm. Field of view of 10 μm .
Quantification of crystalline structure and impurities	Determine if gravity changes the crystalline structure.	Crystal lattice structure, lattice spacing, and impurities.	Electron microscope (Diffraction)	System needs to be able to resolve the structure at sub 0.2 nm resolution.

Mission Requirements
The mission specialist must conduct two distinct experimental phases: one under stable 2G hyper gravity conditions and another during microgravity. At the start of each phase, the laser cavitation system, housed within the interlocked chamber, is activated via the GUI on the control laptop to operate continuously for at least 10 seconds on a designated well of the 384-well plate, which is mounted on a motorized translation stage. After each trial, the stage automatically advances to a new well for duplicate trials, ensuring multiple independent samples for both gravity conditions without manual intervention. Throughout both phases, the accelerometer and camera continuously log synchronized data, with each well's position and corresponding gravity condition recorded for post-flight analysis. The GUI verifies interlock status and operational readiness before each laser activation, and a minimum of three replicates per gravity condition must be completed to ensure statistical validity and reproducibility.

9.0 Software System

The control software is a custom application written in Python that runs on the control laptop and serves as the single point of interaction between the mission specialist and the experimental hardware. The software is organized into three components: the graphical user interface (GUI), a backend hardware abstraction library, and the data files generated during operation. The software for the project has been uploaded to a publicly accessible GitHub page (https://github.com/danielzyu/CRISTAR_UoFT_CAN-RGX_9).

9.1 Graphical User Interface: The GUI is built using the Tkinter framework and provides the mission specialist with all the controls and feedback required to operate the experiment. It is divided into four functional regions: a sensor readout panel displaying real-time acceleration, temperature, and humidity values; a platform control panel for selecting and advancing the 384-well plate sample position via the translation stage; a data logging panel for enabling file output and specifying file paths; and an experiment control panel containing the START EXPERIMENT and FIRE LASER buttons. The live camera feed is displayed to the right of the controls, allowing Operator A to monitor sample ablation in real time. The FIRE LASER button is disabled by default and only becomes active once the experiment has been started, which prevents accidental laser activation outside of nominal operation. A keyboard shortcut (Ctrl+F) is also bound to the fire command for rapid actuation during the limited microgravity window. Prior to each run, the GUI verifies that all interlocks are engaged and that environmental conditions fall within predefined ranges before allowing the laser to be enabled.

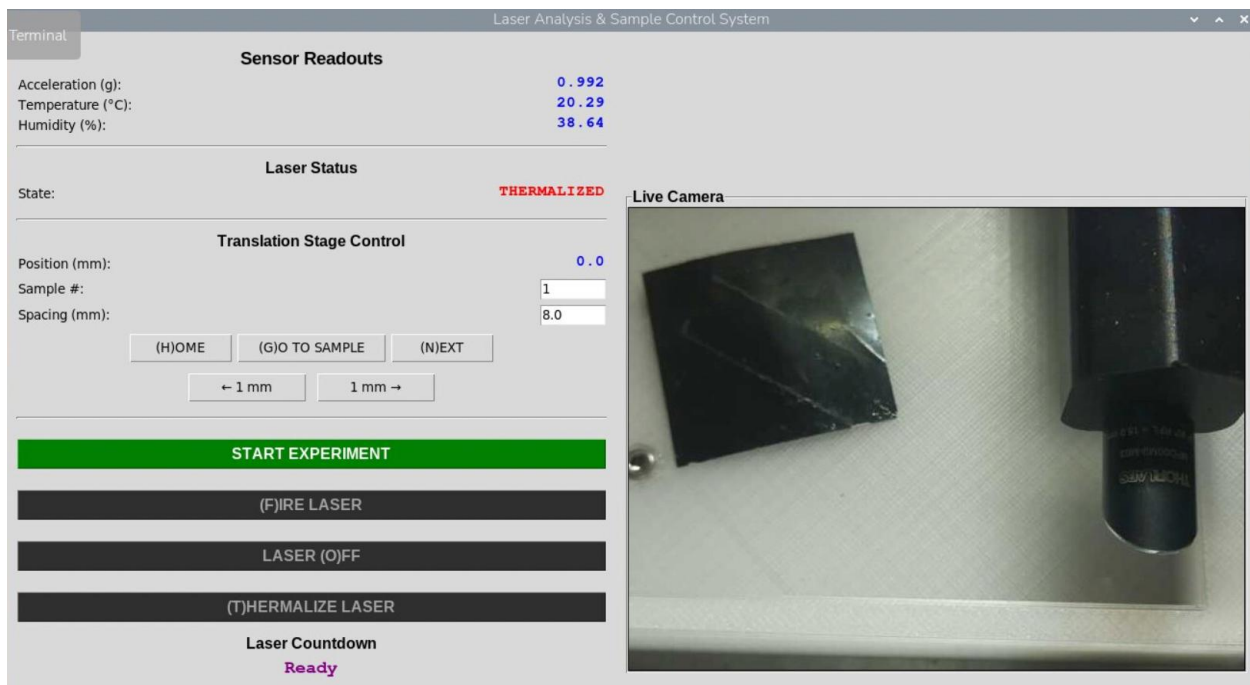


Figure 10: Graphical User Interface screen shot from GUI demonstration video.

9.2 Hardware Library: A custom Python library (`sensors.py`) is imported by the GUI and contains all low-level hardware control logic, including commands to drive the linear translation stage, fire the Ivy laser, poll the BME280 and BNO085 sensors, and stream frames from the camera. The sensor polling, camera streaming, laser and translation stage control modules are fully integrated in the GUI on the raspberry pi. The user communicates with the Raspberry Pi over Ethernet, which in turn interfaces directly with the sensor PCB, translation stage, camera and laser. By isolating hardware-specific code in this library, the GUI remains a clean, high-level controller, and any change to a hardware component (e.g. swapping a sensor) requires modification only to the library and not to the GUI itself. If the library fails to import or a hardware device fails to initialize, the GUI continues to run with the affected readouts held at their default values and the failure logged to the console, allowing the operator to diagnose the issue without the application crashing mid-flight.

9.3 Data Files: When data logging is enabled, the GUI creates a timestamped CSV file at the start of each experiment and appends one row per laser firing event. Each row records the timestamp, sample number, three-axis acceleration, velocity, orientation, temperature, pressure, humidity, and altitude, providing a synchronized record of conditions at the moment of crystal nucleation. Acceleration is reported in units of g, with the conversion from the BNO085's native m/s^2 output performed in software. Camera video is recorded continuously during each parabola and saved as a separate timestamped video file. To satisfy the data segregation requirements of Section 9.2, files generated during hypergravity, microgravity, and level-flight phases are saved into separate folders, each tagged with the parabola number and gravity condition. All data is written to local storage on the control laptop.

9.4 Software Testing: To ensure correct operation of the software control systems, the GUI automatically executes a test program during initialization. This program verifies connectivity to the laser, motion control stage, and sensors before prompting the operator to test keyboard inputs. The testing includes a negative test in which the user is instructed to press random keys to confirm that no system process is inadvertently initiated. The user is also required to press three keys instead of the configured two-key bindings to verify that activation occurs only when exactly two keys are pressed simultaneously. Once these initialization tests are successfully completed, the GUI is considered ready for operator use.

10.0 On-Aircraft procedures

In-flight procedures for the mission specialists are provided below for each phase of flight. A flow chart version of these procedures is provided in Appendix C.

10.1 Personnel and Responsibilities:

Each flight will be conducted with two designated Mission Specialists, identified as Operator A and Operator B, each with clear responsibilities. This division of responsibilities ensures that neither safety monitoring nor system control is left unattended at any point during the flight.

Operator A shall be responsible for:

- GUI Operation
- Laser System Control
- Translation Stage Control
- Camera feedback monitoring
- Data verification

Operator B shall be responsible for:

- Oversight of the physical payload
- Sample Handling
- Monitoring Interlock system and sensors
- Timekeeping throughout each parabolic phase

Both Operators hold equal authority over the emergency stop. The conditions under which the emergency stop shall be engaged are specified in subsection 10.4. All active system control shall be exclusively executed through the custom GUI. The Pelican case shall not be opened during any parabolic manoeuvre until after it is confirmed the system has been fully powered down or the emergency stop has been engaged.

10.2 In Flight Procedures: Nominal Operation

10.2.1 System Initialization

While on the ground before taxi but after power is provided to the cabin, both Operators shall complete the following power-on and interlock verification sequence.

1. **Operator B** shall flip the power switch on the Pelican case to the ON position and confirm that the green LED illuminates, indicating power has been successfully supplied to the system.
2. **Operators A and B** Shall confirm that the Interlock bypass is engaged inside the box.
3. **Operator B** will cycle the key on the back of the laser and ensure that the light on the back of the laser is pulsing green.
4. **Operator B** will then close the box and latch the front two clamps on the pelican case.
5. **Operator B** will remove the interlock bypass and connect the ethernet cable to the box. **Operator A** shall confirm the Raspberry Pi has completed its boot sequence.
6. **Operator A** will then run the “start” in the terminal window. This will run Connection.py code to ensure all systems are operating properly. If the interlock was not initiated properly, return to the start of the initialization procedure (step 1)
7. If the connections are working properly, **Operator A** will then run Cristar_GUI.py by pressing “y” then the enter key. **Operator A** shall confirm GUI responsiveness by checking the system readout for error messages and confirming that the sensors are reporting expected start values (i.e. correct positional index of translation stage, all sensors reading nominal ambient values).
8. **Operator A** will use the GUI to initialize the laser by pressing start experiment. From this point the laser will take 1-5 minutes to stabilize before it can start to emit. This will be indicated by the laser status changing to “Thermalized”. If the system is not thermalized after 5 minutes, the thermalize button will be pressed.
9. After thermalizing, **Operator A** will perform one test firing of the laser in the homed position to ensure the system is ready for microgravity parabolas. A successful laser firing will be when the red enabled light of the laser turns on, as observed from the camera. If the laser does not fire, the laser thermalize for another 10 minutes before next attempt.
10. After successfully testing the laser emission, both operators shall verbally confirm that the system is powered, interlocks are engaged, and ready before the aircraft begins taxi.

10.2.2 Structure of Parabolic Phases

To maximize the length of the hyper gravity window, the preference for the CRISTAR project is continuous parabolas. Sequential individual parabolas can also work for this experiment if

scheduling requires this flight configuration. The ideal flight sequence would be an individual parabola for acclimatisation followed by the sequence of continuous parabolas.

Each data parabola of the continuous sequence comprises of three sequential phases: the first hyper gravity pull, the microgravity window, and the second hyper gravity pull. Each phase has a nominal duration of approximately 20 seconds. Accounting for operator reaction times and transition period timing between actions, effective data-collection windows are estimated to be 10-15 seconds per phase. Passing through 1.5 G, the hyper gravity experiments will commence and be terminated. Passing through 0.5 G, the microgravity experiment will commence and be terminated.

10.2.3 Minimum Parabola Requirement

A minimum of 6 microgravity data collection parabolas are required per flight to satisfy the primary scientific objectives of the experiment. This threshold is representative of the minimum sample size necessary to have statistically meaningful analysis of microgravity-induced effects on the sample material.

10.2.4 Acclimatization Parabola

The first parabola of the flight is an acclimatization parabola. No sample data will be collected. Both Operators shall use this period to orient themselves to the flight environment and verify system readiness prior to the start of experimental data collection.

During the acclimatization parabola, the following verification tasks shall be completed:

- Confirm GUI responsiveness and control functionality
- Verify camera feedback and framing
- Verify sensor feedback and expected range of operation
- Execute and monitor the autonomous aiming sequence without laser activation

10.2.5 Data Collection Parabolas

There are an expected 10 data parabolas. The nominal sequence of operations for each data parabola is specified as below.

10.2.5.1 First Hypergravity Pull

1. **Both Operators** shall issue a verbal ready call, explicitly confirming that all systems and personnel are prepared, prior to entering the hypergravity phase of the parabola. The next steps shall not be initiated until this ready call has been exchanged and acknowledged by both Operators.
2. **Operator A** shall initiate the run sequence via the GUI by pressing “Ctrl+F” once the force of gravity reaches 1.5g.
3. The system shall autonomously execute the following sequence of events: laser activation per preprogrammed parameters, translation stage advancement to the next sample position and automated stop when the edge of the sample is reached. During this time there shall also be continuous camera recording of the internal system, on which **Operator A** shall monitor sample cavitation.
4. No manual intervention shall be permitted during hypergravity conditions beyond authorized emergency override commands (see subsection 10.4)

5. **Operator B** shall monitor the physical state of the payload, note the time the parabola started and monitor the gravity reading on the airplane.
6. After 20 seconds the laser will automatically execute the stop command. In the case where hyper gravity ends before 20 seconds, when gravity returns to 1.5g **Operator A** shall press “Ctrl+F” to disable the laser. The system shall disable the laser, cease camera recording, autosave data to a designated folder with appropriate timestamps and advance to the next sample position.

10.2.5.2 Microgravity Window

1. **Operator B** shall monitor for the onset of microgravity conditions. At 0.5 Operator B will issue a verbal cue to Operator A to start the experiment.
2. **Operator A** shall initiate the run sequence via GUI by pressing “Ctrl+F”. As outlined in step 3 of Section 2.4.1, the automated sequence shall enable the laser, initiate camera recording, advance the translation stage to the sample position and execute laser firing per preprogrammed parameters.
3. **Operator A** shall monitor the live camera feed for laser emission confirmation and bubbling.
4. **Operator B** shall simultaneously time keep and monitor the gravity sensor on the airplane.
5. After 20 seconds the laser will automatically execute the stop command. In the event that the gravity window is shorter than 20 seconds, **Operator B** gives a verbal cue that microgravity has reached 0.5g. **Operator A** will run the stop command (“Ctrl+F”), and the system shall disable the laser, autosave data to a separate folder with timestamps stored separately from the hypergravity datasets, and move the stage to the next position.

10.2.5.3 Second Hyper gravity Pull

In the second hyper gravity pull the operators will follow the same procedure as the first hyper gravity pull.

10.2.6 Level-Flight Stabilization Phase (Inter-Parabolas)

In the event of a need for level flight between parabolas, the following procedure will be used. During the 3-minute phase of level-flight stabilization following a parabola, the system shall be held in a safe standby state with the laser off. The following tasks shall be completed prior to entering the next parabola. The following steps are also equally detailed in Appendix C, Figure 18.

Operator B shall:

1. Confirm the prior run has fully completed and properly terminated.
2. Verify the laser is disabled and the translation stage is stationary.
3. Confirm all data from the preceding run has been labeled and saved appropriately.

Operator A shall:

1. Verify GUI responsiveness.
2. Re-confirm camera framing and feedback stability.
3. Confirm the autonomous aiming sequence is ready.

4. Reset the sample index in the GUI in preparation for the next run.

Laser firing and translation stage motion are prohibited throughout this phase. Within plus or minus 5 seconds from the next parabola as noted by Operator A, both Operators must verbally confirm that all systems are ready to proceed into the next parabolic manoeuvre. This ready call shall be completed prior to entering the hypergravity phase of the upcoming parabola; the parabola shall not be initiated until both Operators have given this confirmation.

10.2.7 Return Transit Flight

After the parabolic sequence is completed, the following procedure will be followed.

1. **Operator A** shall turn off laser by pressing the laser off button on the GUI (or using the shortcut “Ctrl+O”)
2. **Operator A** shall ensure the data is logged and saved
3. **Operator A** will Shutdown the GUI using “sudo poweroff” command in terminal
4. **Operator A** shall open the Pelican case to ensure the raspberry pi powered down
5. **Operator A** shall turn off the power pelican case
6. **Operator B** will confirm that the shutdown procedure has been completed.

10.3 In Flight Procedures: Off -Nominal Operation

10.3.1 Transition Criteria

Mission Specialists shall transition to off-nominal procedures upon the observation of any of the following conditions:

- GUI non-responsiveness or loss of system control.
- Absence of laser emission confirmation on the camera feed following initiation of the sequence.
- Anomalous or absent sample response to laser emission inconsistent to expected ablation behaviour.
- Sensor readings outside acceptable operating procedures.
- Translation stage failure to advance to the designated position.
- Camera feed loss or degraded image quality that hinders monitoring.
- Data save failure or absence of expected output files with correct time stamps.

The following corrective steps are also shown in a procedural flowchart in Appendix C, figure 19.

10.3.2 Off-Nominal Corrective Procedures

Upon transition to an off-nominal status, which shall be verbally confirmed between the mission specialists, the following corrective actions shall be executed as applicable during the level-flight stabilization phase only. If a transition to an off-nominal state occurs during the parabola, the Operators shall disable the laser and turn off the system and wait for the level-flight stabilization phase to continue with the following noted corrective procedures.

1. **Operator A** shall issue a GUI stop command if not already issued to terminate the active sequence. **Operator A** shall confirm the laser is disabled and the translation stage is stationary.

2. **Both Operators** shall jointly assess the nature of the issue and determine whether the system can be restored back to nominal operation within the available inter-parabola window of 3 minutes.
3. **Operator A** shall attempt GUI reset and system re-initialisation as required.
4. **Operator B** shall verify physical payload and sensor status.
5. If the fault is resolved and **both Operators** confirm system readiness, nominal operation shall resume for the next parabola.
6. If the fault cannot be resolved within the next inter-parabola window, the affected parabola shall be designated as void and the issue shall be logged for post-flight review.

10.4 Emergency Shutdown Procedures

10.4.1 Criteria for Emergency Shutdown

Either Operator shall initiate an Emergency Stop upon observation of any of the following conditions.

- Visible smoke, burning odor, or thermal anomaly within or around the Pelican case and interlock
- Structural failure or physical breach of the experiment enclosure or interlock
- Uncontrolled laser emission or evidence of laser light escaping the enclosure
- Electrical fault, sparking, failure or power anomaly
- Any other condition assessed by either Operator or the flight crew that presents an immediate risk to personnel and the aircraft.

10.4.2 Emergency Shutdown Sequence

Either Operator shall audibly declare “Emergency Stop” to alert the flight crew. Either Operator shall immediately engage the hardware Emergency stop by flipping the power to the pelican case off.

1. **Operator A** or B shall confirm the power to the box is disabled by looking at the green LED. If power is off the LED will stop illuminating.
2. **Operator A** shall confirm loss of connection of the computer to Raspberry PI.
3. No further system interaction shall occur until the fault has been assessed and cleared by **both Operators** in coordination with flight crew.

10.4.3 System Response to Power Loss

In the event of a total power loss to the experiment system, the following states shall be assumed.

1. The laser shall default to a disabled state; no emission shall occur without powered system initialisation
2. The translation stage shall stop all motion and hold its last commanded position
3. The camera shall cease recording, data buffered prior to the power loss might not be recoverable
4. The GUI shall become non-responsive- no system commands can be issued without power and until the system is fully re-initialized.

Power restoration requires full system re-initialization and re-verification of all subsystems prior to resuming operation. Any parabola in progress at the time shall be designated as void.

10.4.4 System response to unplugged laser:

If the laser loses connection with the microcontroller, **Operator A** will see an error message appear on the GUI. In this event the following process will be followed:

1. **Operator A and B** will wait for the return to normal gravity and level flight.
2. **Operator A** will shutdown raspberry pi
3. **Operator A** Power off box from outside key
4. **Operator B** will open box, reconnect the laser
5. Follow laser startup procedure.

10.4.5 System response to unplugged microcontroller:

If the microcontroller loses connection with the controlling laptop **operator A** will see the VNC lose connection. In this event the following process will be followed:

1. **Operator A** will check external connections and try to re-connect
2. If external connects are checked and not allowing reconnect, Operators must wait at least 20 seconds.
3. **Operator B** will break the interlock and check internal connections. **Operator A** will then try to re-connect.
4. Continue checking connections until connection is reset.

10.4.6 Laser Interlock Broken:

If the laser interlock is broken the following procedure must be followed:

1. **Operator A** will Shutdown raspberry pi.
2. **Operator A** will power off box from outside key.
3. Follow startup procedure.

11.0 Verification and Compliance Matrix

Full Verification and compliance matrix has been uploaded to the TEDP folder ([link](#)).

12.0 Describe the Structural Verification procedure:

12.1 Physical Testing

PLA printed parts were printed in batches of three. One for destructive load measurement to find ultimate strength. One for non-destructive testing to a load of 3G with repeated cycling of weight on and off for 10 second intervals for 20 seconds each. Finally, one spare part were printed at the same time. If the part needs to be redesigned, the parts were printed again at the same time. Parts was tested using standard weights from the department of Physics at the University of Toronto. If the required weight exceeded the weights available from the standard weights sets, water at set volumes was added to a jug to create the required loads.

All commercially purchased parts were tested with the load of 3G to ensure correct operations under this load. This was accomplished by using a standard of weights for 5-minute load period followed by cycling the weight on and off for five 20 second intervals.

12.1.1 Yield Point Determination of PLA Mounts

Two PLA mirror mounts from the same print batch were tested in a cantilever configuration replicating the mirror with a controlled load (see Figure 11). The base of each mount was clamped to the edge of the baseplate and load was applied vertically at the free end of the cantilever arm via filling a jug with water (jug + lid = 136 g). Deformation was measured as the reduction in vertical distance from the free end of the mount arm to the surface of the baseplate using a digital caliper (± 0.05 mm resolution). As the load was applied via a monofilament fishing line, a small amount of elastic elongation of the line may have contributed to the measured displacement. Consequently, the measured deflection represents a conservative estimate of mount deformation, as any line stretch would slightly overestimate the true deflection of the PLA mount.

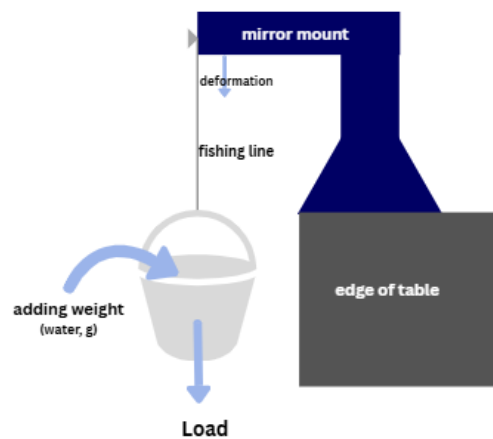


Figure 11: Mirror Mount Setup for Yield Point Determination and Load Testing.

Load was applied incrementally from 136 g to 4136 g total applied load. The functional yield threshold was defined as 0.5 mm deflection from the unloaded reference position, consistent with the alignment tolerance of the optical system. This threshold was first exceeded at a total applied load of 2036 g (19.97 N), resolved well above the ± 0.07 mm uncertainty floor. A linear regression across the full dataset yielded $R^2 = 0.597$ and a stiffness of 42.5 N/mm, confirming elastic behaviour throughout the testing regime (see Figure 12). The operational load on the mirror mount is 63.2 g (0.62 N), which corresponds to the mass of the gold off-axis parabolic mirror under 1G, giving an experimental factor of safety of 44.7. This substantially exceeds the program requirement of 1.5 per RGX-PHY002-A.

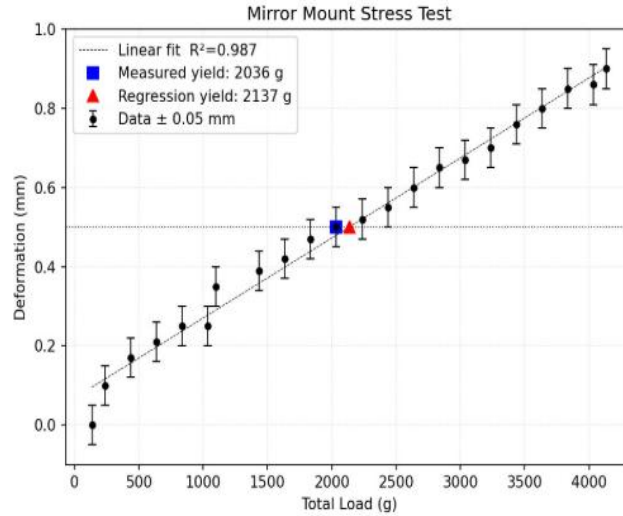


Figure 12: PLA Mirror Mount Deformation under incremental cantilever loading. Dashed line indicates the 0.5mm functional yield point as defined.

12.1.2 Cyclic Loading

Cyclic loading was conducted on two mounts printed in different orientations. One printed paralleled to the base of the mount, and one printed perpendicular to the base of the mount. After unloading after yield testing, both mounts returned to their original positions with no observable permanent deformation. Both mounts were then cycled at the full proof load of 4136 g (40.56 N), comprising 4000 g added mass plus the 136 g jug and lid, for 10 cycles at 20 ± 0.1 seconds per cycle per RGXPHY002-A.

The mount printed parallel to the base of the mount deformed 0.74 mm statically under this load. It was subsequently dropped from 3 inches with the full container attached and failed at the base and was removed from consideration. The mount printed perpendicular to the base deformed 0.91 mm statically under the same load. After 10 cycles the position shifted by 0.01 mm, which is within the ± 0.07 mm measurement uncertainty. No progressive deformation was observed. The perpendicular print did not fail under the subsequent 3-inch drop test with a full container.

The perpendicular printed mount was selected for flight. Static deformation under the 3G proof load is 0.91 ± 0.07 mm with no measurable change after cyclic loading, satisfying RGXPHY002-A.

12.2 Electrical Verification procedure:

A 600 W, 115 V, 5 A uninterrupted power supply was used to power the pelican case for ground testing. This apparatus will also be used for testing complete power loss to show the system defaults into a safe condition. The initial test took place on March 10th, 2026. Our results showed that the laser became disabled and the translation stage locked when the power was cut showing that there are no hazardous conditions created.

12.3 Ground Test procedure:

Ground testing, to the fullest extent possible, followed the same process as the flight procedure. ([The ground test video is linked here for review.](#)) Operator B was responsible for overseeing the

physical payload, sample handling, monitoring the interlock, sensors, and timing. Operator A handled all GUI operations, which included laser control, translation stage control, camera feedback monitoring, and data verification. Prior to the ground test, the system was inspected by both operators to ensure the configuration matched flight operations.

Prior to the start of the ground test, the 384-well plate was prepared and placed in the box. The ground test was conducted in a laser-safe laboratory, with both operators having completed the University of Toronto Laser Safety Course. To simulate operations during the parabolas, a laptop was placed on top of the Pelican case, and the top holes were covered with electrical tape.

To simulate the first parabola for data collection, Operator B confirmed that the experiment was properly set up, carefully noting stage position, laser readiness, and sample positioning. Operator A confirmed that the GUI and its controls were responsive, the camera framing and feedback were ready, and the sensors were functioning as expected.

Both operators verbally agreed to proceed before Operator A initiated the run sequence with the laser enabled in the GUI. The laser fired within the Pelican case according to preprogrammed parameters, the translation stage autonomously advanced to the designated sample position, and the camera feed recorded continuously to monitor both the internal system and sample ablation. Operator B monitored the physical state of the payload, maintained timekeeping, and tracked the sensors. Both operators verbally confirmed readiness, Operator A initiated the run through the GUI, and the laser firing, autonomous stage movement, and camera recording sequence proceeded identically. No manual intervention occurred during the simulated "hypergravity" conditions, which were modeled by a 20-second delay during which the operators could not intervene in the Pelican case's operation.

After the 20-second pause, Operator B monitored the gravity cue (orally confirmed during the ground test) and, within ± 2 seconds of "confirmed microgravity," verbally cued Operator A to press "Laser Fire" in the GUI. The simulated microgravity window lasted approximately 20 seconds. The automated sequence enabled the laser, started camera recording, and commanded the stage to advance to the current sample position, where the laser fired according to the preprogrammed exposure parameters. Operator A monitored the live camera feed for confirmation of laser emission and visible sample response, such as bubbling, while Operator B simultaneously monitored the physical payload. Upon completion, the laser was disabled, stage motion ceased, and data were automatically saved to a specified folder on the computer with timestamps.

To simulate the conditions that would occur if the microgravity and hyper gravity windows were shorter than 20 seconds, the tests were completed again with the operator manually ending the irradiation process using the "stop" input on the GUI.

The second simulated hypergravity (2G) pull was represented by another 20-second pause, during which the operators could not interact with the experiment.

The ground test was considered successful when all wells were successfully irradiated by the laser and crystals were successfully harvested from the well plate.

12.4 Additional Verification:

As our system is laser based, we demonstrated that the enclosure is both light tight and that the laser cannot be active if the box is open. Details on laser safety calculations are provided in Appendix B. To facilitate this testing the following two verifications are required:

12.4.1 Dark room testing of the light tight box:

To show that the box is light tight, the box was placed in a dark room in the department of physics at the University of Toronto. While in the dark room, a bright LED array was placed inside the box with the lid opened. The room lights will then be turned off. A video was then be started showing the bright light in the box. With the video recording, the box was closed, and a white sheet of paper was moved around the box to look for any escaping light. This test successfully showed that no light escaped the box. ([Video link](#))

12.4.2 Laser interlock testing:

To show that the interlock system works, we will first demonstrated on the manufacturer's GUI that the interlock is triggered when the box is open or closed. Then, in a laser safe lab, we will enable the laser with the box closed, with the laser illuminating a Photodiode sensitive to the Mid-IR. With the box opened during laser operations we will check using an oscilloscope that the laser did cease operations when the box was open and did not restart operations when the box closed until the user initiated the start of the laser. After successfully showing that the interlock worked, the following video was taken to show the interlock in action. ([Video link](#))

13.0 Hazard Analysis Report

Efforts have been made to minimize the risks and hazards associated with the CRISTAR experiment. However, some residual risks remain. The hazards present in this experiment can be broadly categorized into three groups: chemical, electrical, and optical. Details of each hazard category and the associated mitigation strategies are summarized in Tables 5-10 below.

Chemical hazards can include exposure to corrosive substances, toxins, or other hazardous reagents. In this experiment, samples are prepared using non-hazardous chemicals wherever possible. The primary sample being crystallized is hen egg white lysozyme, which is non-toxic and generally considered non-hazardous under standard laboratory handling conditions. The sample is dissolved in a salt solution that is also non-toxic and non-hazardous in nature. Relevant Safety Data Sheets (SDS) for all chemicals used in the experiment are provided in Appendix D. Details on the major identified hazard of a sample spill are outline in Table 5.

Table 5: Protein Crystallization solution spills.

Hazard: Sample solution spills	
Description of Hazard to Scientific Outcome	Description of Hazard to Crew or Aircraft Safety
Experiments cannot be performed if the sample is not contained.	
Description of Hazard Control and Mitigation Plan	
Possibility of the solution leaving the enclosed sample container during flight. This is mitigated by using non-hazardous samples, chicken egg white lysozyme and glycine (see SDS sheets in Appendix), having full enclosure inside the box and having a sample that is water soluble to protect equipment.	
Cease Test Criteria:	
If the solution is seen outside the container on the camera, stop the test.	
Terminate Flight Criteria:	
NA	

Electrical hazards include the potential for overload, short circuits, and heating of electrical components. These risks are mitigated by operating all electrical units well below their maximum current and voltage ratings. Additionally, proper wiring practices and rated components are used to ensure safe operation throughout the experiment. Details on these hazards are listed in Tables 6 and 7.

Table 6: Power loss hazard table.

Hazard: Loss of power	
Description of Hazard to Scientific Outcome	Description of Hazard to Crew or Aircraft Safety
If power to the system is lost mid-flight, the experiment cannot be conducted.	NA
Description of Hazard Control and Mitigation Plan	
If the power cable gets disconnected during flight or if the internal power bar comes disconnected the experiment will not operate.	
Cease Test Criteria:	
If power is lost during a test, cease the test and identify the loss of power. If the green light on the box is on, cease the test. If the green light on the box is off. Look at the plug to identify if it is plugged in. Cycle the power switch to see if power is restored. If power is restored, continue with the next parabola.	
Terminate Flight Criteria:	
NA	

Table 7: Translation stage failure hazard table.

Hazard: Translation stage failure	
Description of Hazard to Scientific Outcome	Description of Hazard to Crew or Aircraft Safety
If the translation stage fails only one sample can be produced per flight	NA
Description of Hazard Control and Mitigation Plan	
The selected translation stage is durable and can be manual operated. Prior to the flight the translation stage will be tested to ensure functionality. If the stage becomes inoperable during the flight, the stage should be reset and tested again.	
Cease Test Criteria:	
Stage is unable to move. Cease test.	
Terminate Flight Criteria:	
NA	

Optical hazards arise from the use of laser radiation within the experimental system. These risks are mitigated through the implementation of appropriate safety measures, including interlock systems, controlled beam paths, and adherence to established laser safety protocols. These measures are intended to prevent accidental exposure to laser radiation during operation. Details on these hazards are listed in Tables 8-10.

Table 8: Loss of Connections with laser.

Hazard: Loss of Connections with laser.	
Description of Hazard to Scientific Outcome	Description of Hazard to Crew or Aircraft Safety
Unable to stop the laser from irradiating samples adds uncertainty to laser parameters used on the samples.	Unable to determine if the laser is operating can pose a laser safety risk.
Description of Hazard Control and Mitigation Plan	
If connection is lost to the laser when the laser is operating this poses an experimental hazard as the irradiation time of the sample will not be known and a laser safety hazard as the condition of the laser is unknown. To mitigate the risk, if the laser connection is lost power to the box must be cut off, ceasing the experiment.	
Cease Test Criteria:	
If the laser connection is lost power to the box must be cut off, ceasing the experiment.	
Terminate Flight Criteria:	
NA	

Table 9: Structural failure or deformation of the Mirror Mount.

Hazard: Structural failure or deformation of the Mirror Mount	
Description of Hazard to Scientific Outcome	Description of Hazard to Crew or Aircraft Safety
Deformation of the mirror mount during flight may cause laser light to not hit the intended location on the well plate.	NA
Description of Hazard Control and Mitigation Plan	
Mirror mount changes location by more than 0.5 mm. the mirror mount has a much higher factor of safety (over 50) than other components. As such, mirror deformation should not occur.	
Cease Test Criteria:	
If the mirror mount is seen deforming on the camera and cavitation ceases prematurely, end the test.	
Terminate Flight Criteria:	
NA	

Table 10: Loss of laser light confinement risk table.

Hazard: Loss of laser light confinement	
Description of Hazard to Scientific Outcome	Description of Hazard to Crew or Aircraft Safety
Cavitation would cease, preventing crystal formation.	Potential for laser injuries to crew if the laser light is allowed to escape the confinement of the pelican case.
Description of Hazard Control and Mitigation Plan	
Possibility for laser light to escape the optical set up, and the pelican case enclosure. This is mitigated through making the box light tight, use of fixed optics, non-reflective surfaces in the box and the use of interlocks to prevent laser operations if the box is open. During level flight before and after the parabolas the laser system will be disabled. Under normal operation, the enclosure and safety interlocks ensure the system operates as a Class 1 laser product and laser safety eyewear is not required. Operators will be provided with OD5+ laser safety goggles for use only during servicing, alignment, or other procedures where the enclosure is open or safety interlocks are intentionally defeated, as described in Appendix B. These procedures will not occur after departing UofT as they require a laser safe lab.	
Cease Test Criteria:	
If the interlocks are triggered or the internal mirror is misaligned, cease the test. If the light is seen escaping the box, testing will cease.	
Terminate Flight Criteria:	
NA	

14.0 Ground Support and Crew Assistance

14.1 Specialised Equipment:

To preserve the HEWL and glycine crystallization samples formed during the microgravity experiments, a room temperature storage approach at 20°C will be maintained throughout transport and handling. To this end, we intend to bring a Styrofoam container as soon as possible following the flight. To this end, a Styrofoam container will be used at the campaign site to store the 384-well plates both prior to and following flight operations. All samples will remain sealed within the closed 384-well plates to prevent evaporation and unintended crystallization effects, which is common for HEWL and glycine samples. The well plates will be prepared in advance at the University of Toronto and transported to the campaign site. A small dino-lite microscope will be used to look at the wells immediately after the flight to see if crystals formed.

14.2 Ground Crew Procedures:

14.2.1 Prior to arrival at Campaign site: The ground crew will be responsible for preparing the samples prior to departure from the University of Toronto. The time when the solution and supernatant are mixed will be noted. The ground crew will be responsible for the transportation of samples to and from the campaign site.

14.2.2 Prior to Flight: Samples will be brought to the ground crew by the flight crew and loaded by the flight crew and preflight testing will be conducted by the flight crew. The time of mixing will be reported to the flight and ground crew (accurate to the nearest minute).

14.2.3 Post Flight: The ground crew will collect the well plate from the flight crew, ensuring the samples are appropriately sealed, and place them into the Styrofoam container for transportation. Upon return to the lab, samples will be immediately cryo-preserved on SEM grids using a Leica GP2.

Note: Samples should be collected from the 384 well plate within 1 week of production. The impact of immediate and delayed extraction from the 384 well was investigated and it was determined that the crystal structures and solution would be stable for up to seven days.

15.0 Technical Risk Assessment

Potential points of failure for the project are outlined in detail in Appendix A. The summary of the risk assessment table is shown in the risk assessment matrix below (Table 11). We note that most of the failure modes that would result in an unsuccessful experiment are technical in nature. These risk assessment points are mitigated through a combination of engineering design and ensuring replacement parts for critical components are available at the campaign site.

Table 11: Risk Assessment Matrix.

		Probability		
		Low	Med	High
Severity	High	MR1, TR1, TR2, TR3, TR4		
	Med	HR2, HR3, TR6	MR2, MR3	
	Low	TR5		HR1

16.0 Project Management

16.1 Team structure and Management

CRISTAR is a team of seven students at the University of Toronto, with one faculty advisor. The team consists of two graduate students from the department of physics, and five undergraduate students from the Department of Engineering. To ensure the successful completion of our project objectives, tasks have been distributed as outlined in the work breakdown structure in Figure 13, with role assignment being given in Table 12. Roles were allocated according to areas of expertise and interest. We decided to have concrete roles to make job alignment and accountability straightforward. However, everyone is expected to help outside of their role when needed since interdisciplinary collaboration is at the centre of the team ethos. The work breakdown structure in Figure 13 is written in terms of lines of effort instead of the five project phases to better illustrate the team workload distribution.

Table 12: List of team members, including the faculty advisor, all from the University of Toronto, with roles identified.

First, Last Name	Degree in progress	Level of Study	Department	Role	Email
Alexander, Wainwright	Ph.D.	Graduate	Physics	Team Lead	Alexander.Wainwright@mail.utoronto.ca
Syeda, Mahdia	BASc.	Undergraduate	Engineering Sciences	Science Lead	Syeda.Mahdia @mail.utoronto.ca
Lauren, Altomare	BASc.	Undergraduate	Engineering Sciences	Systems Integration	Lauren.Altomare@mail.utoronto.ca
Daniel, Yu	BASc.	Undergraduate	Computer Engineering	Electrical and Hardware Support	Danielz.Yu@mail.utoronto.ca
Elias, Barsa	BASc.	Undergraduate	Engineering Sciences	Hardware Lead	Elias.Barsa@mail.utoronto.ca
Gabriel, Caribe	BASc.	Undergraduate	Engineering Sciences	Electrical Lead	G.Caribe@mail.utoronto.ca
Natalie Djuric	BSc	Undergraduate	Astro Physics	Outreach	n.djuric@mail.utoronto.ca
Ian Mo	BSc	Undergraduate	Physics	Outreach Support	ian.mo@mail.utoronto.ca
Khaled, Madhoun	Ph.D.	Graduate	Physics	Mentor	Khaled.Madhoun@mail.utoronto.ca
R.J. Dwayne, Miller	-	University Professor	Physics & Chemistry	Faculty Advisor	Dwayne.Miller@utoronto.ca

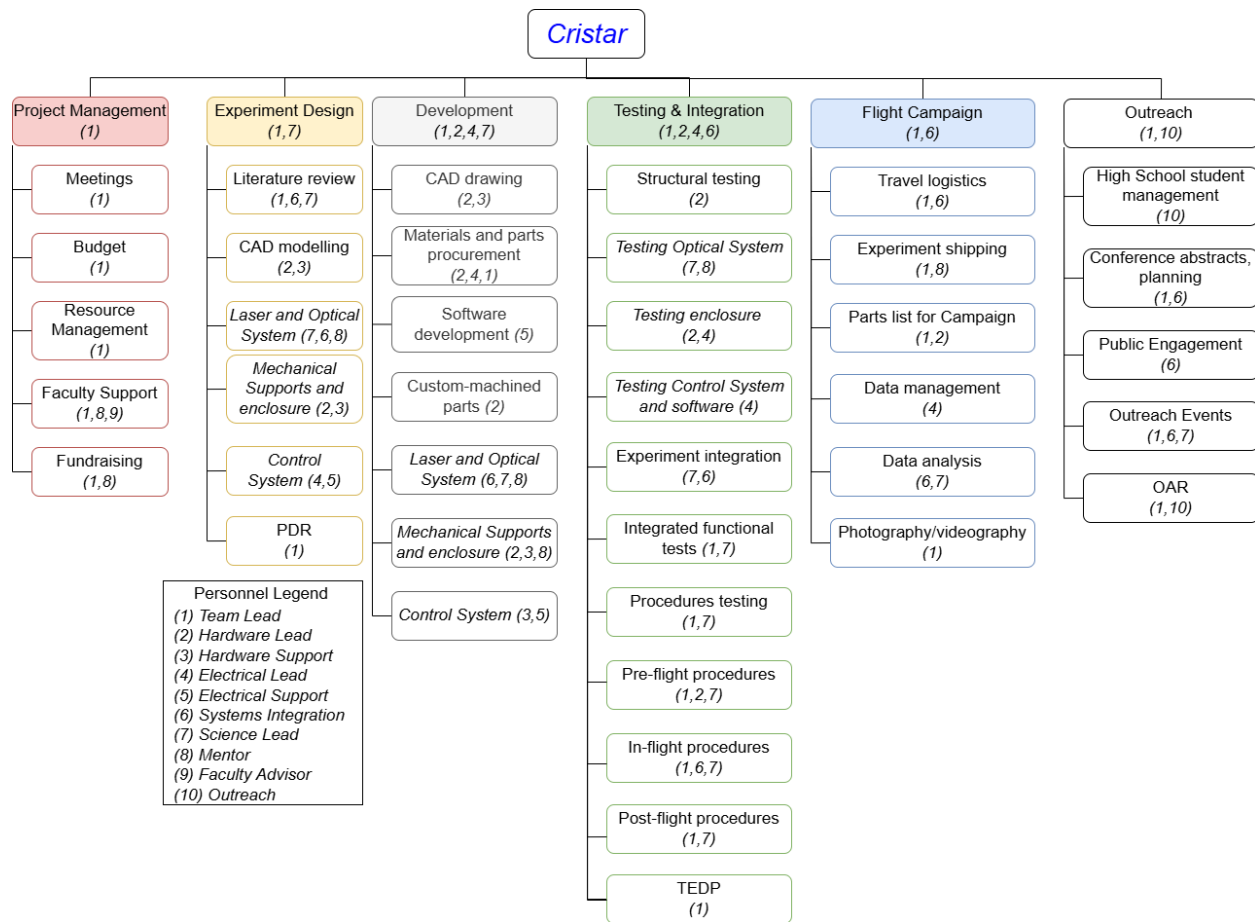


Figure 13: Work breakdown structure for project tasks and responsibilities for each member.

16.2 Project Time Allocation

Given the short timeline and demanding engineering requirements, the project was divided into five distinct phases: design development, ground testing, pre-flight campaign, flight campaign, and post-flight review. During the design development phase, we will create detailed concepts for each component of the microgravity system. Each component will first undergo independent testing to ensure compliance with program requirements and scientific objectives. Once verified, the components were integrated for ground testing, where we assessed the overall system performance against engineering standards for flight safety, laser safety, and experimental reliability. Following ground testing, the pre-flight campaign phase will simulate in-flight procedures on the ground to validate operational readiness. The system will then be deployed for microgravity flights and experiment execution. Finally, in the post-flight review phase, we will analyze results, publish findings, and support ongoing public outreach. In parallel with these engineering objectives, we will conduct both in-person and digital outreach programs to engage the broader community. Details as to task and time allocations for the project are given in Tables 16 to 19 below.

Table 13: Preliminary Design Schedule.

TASK	ASSIGNED	PROGRESS	START	DAYS	END
Preliminary Design Development		100.00%	25/2/2026		30/4/2026
Structure and Organization Development	Alex	100%	25/2/2026	6	2/3/2026
Finish Verification and Compliance Matrix	Alex	100%	2/3/2026	8	9/3/2026
Preliminary Design Development (Optical system)	Elias, Gabriel	100%	9/3/2026	15	23/3/2026
Preliminary Design Development (Science)	Alex	100%	9/3/2026	24	1/4/2026
Preliminary Design Development (Electrical)	Gabriel	100%	9/3/2026	24	1/4/2026
Preliminary Design Development (Hardware)	Elias	100%	9/3/2026	24	1/4/2026
Preliminary Design evaluation and selection (Optical system)	Elias, Gabriel	100%	5/1/2026	2	6/1/2026
Preliminary Design evaluation and selection (Samples)	Alex & Lauren	100%	5/1/2026	12	16/1/2026
Preliminary Design evaluation and selection (Electrical)	Gabriel & Daniel	100%	5/1/2026	12	16/1/2026
Preliminary Design evaluation and selection (Mechanical)	Elias & Daniel	100%	5/1/2026	12	16/1/2026
Flight Procedure Development	Syeda & Lauren	100%	25/2/2026	42	7/4/2026
Ground Test Plan Development	Elias, Gabriel, Alex	100%	1/4/2026	30	30/4/2026
Check design with SMEs (Particular attention to fasteners, loads, and laser safety)	Alex	100%	16/1/2026	7	22/1/2026
Preliminary Design stress test on CAD (Mechanical)	100	100%	5/1/2026	30	3/2/2026
Budget and Fundraising Finalization	Alex	100%	16/1/2026	7	22/1/2026
Outreach Plan Finalization	Natalie	100%	25/2/2026	60	25/4/2026

Table 14: Ground test schedule.

TASK	ASSIGNED	PROGRESS	START	DAYS	END
Ground Testing		100.00%	16/1/2026		14/3/2026
Order parts	Alex	100%	16/1/2026	2	17/1/2026
Optical System and interlock Prototype	Khaled	100%	16/1/2026	14	29/1/2026
Prototype Crystal generation	Alex	100%	16/1/2026	30	14/2/2026
Prototype electrical design	Daniel	100%	16/1/2026	30	14/2/2026
Prototype Mechanical design	Elias	100%	16/1/2026	30	14/2/2026
Stress test Mechanical Design	Elias	100%	14/2/2026	7	20/2/2026
Stress test integrated Electrical Design	Khaled	100%	14/2/2026	7	20/2/2026
Preliminary Design Review	Alex	100%	20/2/2026	4	23/2/2026
Integrate components	Elias	100%	20/2/2026	7	26/2/2026
Stress test total system	Elias	100%	26/2/2026	14	11/3/2026
PDR 1 and Test Equipment Data Package Rev 2	Alex	100%	11/3/2026	4	14/3/2026

Table 15: Preflight testing schedule.

TASK	ASSIGNED	PROGRESS	START	DAYS	END
Preflight Testing		97.22%	20/2/2026		14/8/2026
Full ground test 1	Elias	100%	20/2/2026	30	21/3/2026
Design iteration 1	Elias	100%	21/3/2026	14	3/4/2026
Progress Presentation 2 (PP3)	Elias & Gabriel	100%	23/3/2026	5	27/3/2026
Full ground test 2	Alex	100%	21/3/2026	7	27/3/2026
Design iteration 2	Alex	100%	27/3/2026	21	16/4/2026
Critical Design Review (CDR)	Alex	100%	13/4/2026	7	19/4/2026
Full system integrated experiments	Alex	100%	13/4/2026	37	19/5/2026
Design iteration 3	Alex	100%	19/5/2026	21	9/6/2026
Progress Presentation 3 (PP3)	Natalie	100%	1/6/2026	7	7/6/2026
Full system integrated experiments	Alex	100%	9/6/2026	14	22/6/2026
Ground test video prep.	Natalie	100%	22/6/2026	7	29/6/2026
Prepare flight checklists	Syeda	100%	9/6/2026	14	23/6/2026
Submit Ground Test Video (GTV)	Syeda	100%	7/6/2026	48	24/7/2026
Modify the well plate cover to 384 well plate	Alex	100%	24/7/2026	14	7/8/2026
Flight Readiness Review (FRR)	Alex	100%	4/8/2026	4	7/8/2026
Finalize checklists	Syeda	100%	24/7/2026	15	7/8/2026
XRD crystallization evaluation	Alex	50%	24/7/2026	20	13/8/2026
Submit final TEDP, Rev C	Alex	100%	7/8/2026	8	14/8/2026

Table 16: Flight test schedule.

TASK	ASSIGNED	PROGRESS	START	DAYS	END
Flight Test			31/8/2026		6/9/2026
Flight Campaign	Alex	0%	31/8/2026	7	6/9/2026
Diffraction Imaging	Alex	0%	6/9/2026	30	5/10/2026
Post Flight Review			6/9/2026		27/10/2026

17.0 Budget and Funding

Our funding strategy focused on three key approaches: Connecting, Applying, and Collaborating. First, we connected with corporations to secure “in-kind” support by requesting optical and mechanical components and seeking discounts where donations are not possible. Second, we applied for financial assistance through university funding programs and student organizational funding opportunities to ensure adequate resources for the project. Finally, we collaborated with industry stakeholders by requesting specialized support. These combined efforts helped us secure both material and financial resources while fostering strategic partnerships. Beyond these efforts, CRISTAR was able to borrow materials from the Miller Lab in the departments of physics and chemistry at the University of Toronto, which substantially reduced costs. An outline of the total expenditure for the project is given in Table 21. An overview of our funding requested and received is given in Table 17.

Table 18

Table 17: Budget requirements for project components are travel costs are estimated and described. Shipping costs are planned to be covered by the Miller Lab and are not included in the budget below. Final costs for travel to the travel campaign have not been accumulated to date, resulting in the use of the estimated costs (~) for the total expenditure for the CRISTAR project.

Item	Source	Estimated Cost	Actual Cost
Translation Stages	Zaber	\$150.00	\$3695.00
Camera	Digi Key	\$500.00	0.00 (Miller Lab)
Focusing Lens	Thorlabs	\$250.00	\$284.00
Camera Mount	Thorlabs	\$300.00	\$0.00 (Miller Lab)
Electrical Components	Digi Key	\$300.00	\$150.00
Interlock system	Kentek	\$50.00	\$275.00
Enclosure materials	Kentek	\$500.00	\$0.00 (Miller Lab)
Metal for Fixed Optics	McMaster-Carr	\$500.00	\$0.00 (Miller Lab)
Travel Costs	SPIE/Optica	\$3000.00	~\$3000.00
Total Cost		5750.00	~\$7404.00

Table 18: Funding Sources are listed with green representing secured funds with total expenditure in red.

Funding Source	Confirmed (Y/N)	Amount Expected	Amount Received
SPIE Student Chapter	Yes	\$1000	\$1000.00
Humans Learning Machine Learning (Physics Department)	Yes	\$500	\$500.00
Zaber	Yes	\$2586.5	\$2586.50
Miller Lab	Yes	\$1500.00	\$1534.91
CSA individual travel grant	Yes	\$2000.00	\$3200
Total Funding		\$ 7586.5	\$8786.50
Costs		~\$7404.00	~\$7404.00
Remaining Funds		~\$182.50	~\$1382.5

18.0 Outreach Activity Report

The digital outreach plan to use the social media platforms Instagram and LinkedIn has allowed us to share our work, promote the team, and explain the science behind the project. Additionally, a CRISTAR page currently exists on the University of Toronto Photonics Innovation Center website. This page serves as a log of our progress, and a central hub for information about the project.

Instagram is one of the most widely used social media platforms by the general public, especially among youth. It is casual and versatile, with many forms of uploads possible. We have used our Instagram page to make meaningful and digestible content about our project, such as short form videos and photographs of our work, and interviews with members of our engineering design team (see *Figure 14*). We have linked our GitHub repository in the biography, as well as our new website, with the goal of making scientific knowledge more publicly accessible in every regard. Furthermore, we plan to document our flight and campaign experiences in depth on the platform. Aside from the intended audience, users with prior relationships to our group members boost our visibility simply by interacting with our content. As such, we have tripled our follower count over the past summer, and we are currently seeing a consistent increase in view count on subsequent media posts, indicating successful outreach strategies favouring the social engagement algorithm.

Our in-person outreach plan to network with high schools and youth programs has been somewhat successful, but there were less openings for us than expected. We emailed various people and institutions for the chance to share our work; public and private school boards in both British Columbia and Ontario, science content creators on YouTube, and the SickKids hospital were some of the groups we targeted. Presentations about our work, demonstrations replicating

various parts of the experiment, and conversations about STEM would form the basis of our visits to these institutions. We emphasized teaching students and other people new concepts and bringing them one step closer to hands-on research. We also hoped to foster curiosity and answer questions about our work, thereby inspiring scientifically inclined members of the community at any level of understanding. Some issues we faced along the way were a lack of time in other organizations' schedules for us, as well as policy changes within certain institutions about visitors; people we corresponded with frequently cited a busy summer season. In the fall, Natalie plans to go to Laura Secord Secondary School of the District School Board of Niagara to give a presentation about Team CRISTAR's project and campaign journey to students.

On Saturday, March 28th, Natalie, Gabriel, and Syeda attended the University of Toronto Engineering Conference for High Schoolers (UTECHS) from 13:00-14:00, occupying a room in Bahen Center for Information Technology with props and demonstrations (Figure 15). Adolescent students participating in the conference and touring the different events visited the CRISTAR room, asking questions about the project and its significance, and about engineering at UofT in general. Discussions with David Manly from the SickKids hospital are still underway as the summertime is busy for them, so it may be possible to share our work with them too, but this will have to occur past the duration of this project with SEDS.

On May 9th, Science Rendezvous hosted Team CRISTAR and allotted a booth for our Pelican case. This event was attended by adults and youths, lasting from early morning to 17:00. We consider this event to be a major success due to high attendance, constant traffic around our booth, and a diverse demographic of attendees.

A new team member, fellow UofT undergraduate student Ian Mo, has joined the team to help with outreach efforts. We plan to continue frequent social media activity with the help of his digital media skills. Since his addition to the team, we have developed and launched a website to continue to expand our community outreach online and provide a central hub for progress updates, forms of contact, media links, and sharing scientific knowledge (Figure 16).

Team CRISTAR's work was featured in the 2026 Optics Summer School Program (OSSP) on August 8th as part of the photonics institute lab demonstration (Figure 17). OSSP is an event geared toward undergraduate students of physics and engineering. The 3-day program focuses on optics and optical instrumentation in laboratories as well as quantum concepts surrounding optical physics and lasers. This year's turnout was an unprecedented success, and the many students in attendance were most strongly captivated by the CRISTAR project on display, sparking an increase in social media engagement.

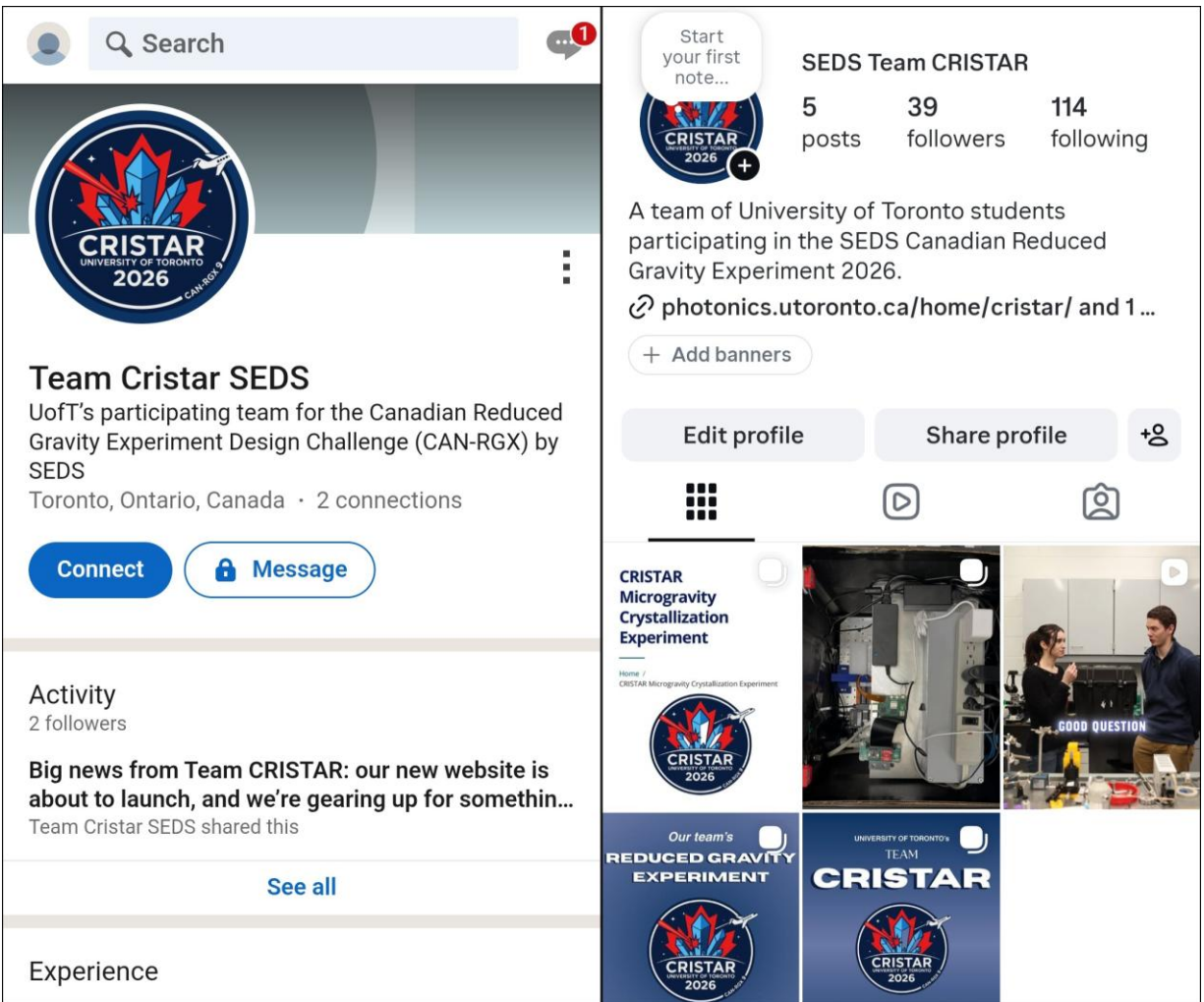


Figure 14: Outreach social media accounts: Left) LinkedIn profile, Right) Instagram account.



Figure 15: CRISTAR's presentation at the 2026 UTECHS Conference, Bahen Centre.

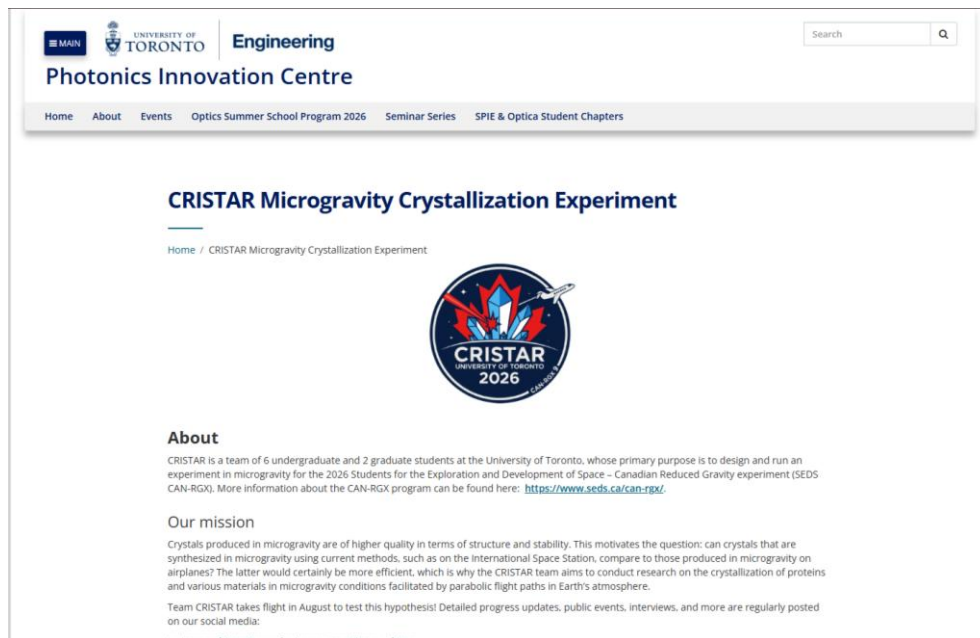


Figure 16: CRISTAR website, hosted as a subsection of the UofT Photonics Innovation Centre page.



Figure 17: Demonstrations at the 2026 OSSP Lab Tour, Lash Miller Chemical Laboratories.

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Appendix A: Risk Assessment Tables

A 1.0 Human Risks (HRs):

Table 19: Team member unavailability Risk Assessment Table.

Risk Event HR1	Team member unavailability during flight campaign due to illness, or personnel commitments.	
Probability	H	Planning over three months in advance can be difficult for undergraduate students who are applying for summer jobs. As such there is a reasonable possibility that one or more members of the team will not be able to attend the flight campaign.
Severity	L	Changing members task allocation based on availability changes will have a low impact on the project as we have cross trained members and a surplus of members available for the project (minimum needed is 3 members, we have 6 at the campaign with current planning).
Mitigation Plan	Passive: Task reallocation protocol, cross-training, and assigned backup personnel to each mission critical tasking. This way, if a team member cannot attend the flight phase of the project the project will be able to continue with minimal disruption.	
Contingency Plan	Reassign tasks; hire replacement; adjust timeline.*	

Table 20: Errors in experimental setup or execution due to human mistakes risk assessment table.

Risk Event HR2	Errors in experimental setup or execution due to human mistakes	
Probability	L	By adopting methods from good manufacturing practices, the risk of human error in experimental setup can be significantly reduced.
Severity	M	If samples are prepared incorrectly or improper experimental execution occurs during flight operations the experimental results will not be valid.
Mitigation Plan	Active: Clear SOPs, peer review of setup, pre-flight simulations will reduce operator mistakes. Specifically, adopting an operator and supervisor method where the operator performs a task and a supervisor monitors the task will reduce errors.	
Contingency Plan	Preparing multiple (three) replicates will allow for multiple experiments to be performed if a mistake is made. Results of significance can still be found.	

Table 21: Inadequate knowledge transfer.

Risk Event HR3	Team members are unaware of mission critical information resulting in mistakes or variations in experimental execution between runs.	
Probability	L	Students have access to shared documentation and will conduct multiple rounds of ground testing to ensure all crew members have adequate knowledge of the experiment and procedure.
Severity	M	If crew members are unaware of mission or experiment critical information they may make mistakes in the SOPs.
Mitigation Plan	Passive: All experimental documents are available to all team members on the shared drive. Active: All crew members will conduct a minimum of two (2) successful ground experiments before the flight, with the primary and secondary flight crews doing a minimum of four (4) full rehearsals before the flight. This way all primary and backup flight crew members are familiar with the flight and ground procedures before going to the campaign site. A flight SOP will also be printed and brought onboard for the flight tests.	
Contingency Plan	If a crew member has inadequate knowledge on the flight, they will have sufficient information available in the SOP to make the best-informed decision possible. If the unknown information is with regards to safety the power to the pelican case will be turned off and experiment terminated.	

A 2.0 Managerial Risks (MRs):

Table 22: Transportation to Campaign site risk assessment table.

Risk Event – MR1	Damage to apparatus from transportation to campaign site.	
Probability	L	Rough handling of equipment when transporting to the campaign site could result in damage to experimental components.
Severity	H	If critical components are damaged, the experiment may not be possible.
Mitigation Plan	The laser will be transported separately from the pelican case to the campaign site in its protective case. As all other components have backups which can easily be swapped out, this reduces the risk of critical damage.	
Contingency Plan	If components are damaged during shipping, they will be replaced with a spare part. To facilitate this, one spare will be available for each component.	

Table 23: Schedule slips due to unforeseen technical or administrative delays.

Risk Event – MR2	Schedule slips due to unforeseen technical or administrative delays	
Probability	M	During design testing, integration and iteration there is a significant possibility that parts will have to be re-designed or replacement parts ordered.
Severity	M	Delays in replacement parts or re-designs can result in delays in reaching CDR and preflight testing goals.
Mitigation Plan	A two-week buffer period has been built in between the PDR and CDR to allow for testing and redesign of critical components based on testing. We also have extra critical components available (focusing mirrors, raspberry pi controllers, etc.) to prevent the need for ordering additional components.	
Contingency Plan	If there are problems integrating the controller and electronic components between the PDR and CDR, the CDR can be completed with manual inputs to the laser and control stage a minimal viable product. This gives more time to fix the integration or order required parts before the campaign phase.	

Table 24: Inefficient task allocation or overloaded team members

Risk Event – MR3	Inefficient task allocation or overloaded team members	
Probability	M	As the team approaches final exam season it is possible that members will not be able to meet their team commitments.
Severity	M	Poor time and task allocation of the team will result in not reaching flight clearance.
Mitigation Plan	Our project planning has been front loaded, limiting the number of tasks required during the summer.	
Contingency Plan	If timings slip and more tasks are pushed to the end of summer, tasks will be reallocated on or before July 20 th , to ensure the team can meet FRR objectives.	

A 3.0 Technical Risks (TRs):

Table 25: Spillage or loss of samples

Risk Event – TR1	Spillage/loss of samples	
Probability	L	Using a fully contained well plate, the chance of spilling samples is low.
Severity	H	If spillage does occur, then the experiment would terminate.
Mitigation Plan	Using a double lidded system, the probability of leakage is minimized.	
Contingency Plan	If the sample holder is found to leak during inverted testing a more viscous solution can be used as the supernatant including agarose and glycerol. These more viscous solutions will reduce the probability of leakage.	

Table 26: Loss of optical alignment.

Risk Event – TR2	Loss of optical alignment	
Probability	L	Using an off axis parabolic mirror with a fixed laser and mount minimizes the possibility of losing laser assignment.
Severity	H	If the laser loses alignment, cavitation will not be possible, preventing a successful experiment.
Mitigation Plan	Using a fixed off axis parabolic mirror reduces alignment challenges as the mirror will focus light to the same location if the light hits the mirror at a near perpendicular angle. As such, the alignment tolerance is approximately 0.5 mm.	
Contingency Plan	If the laser loses alignment prior to the flight, the position of the mirror will need to be modified by replacing the mirror mount with a substitute.	

Table 27: Crystal cracking or No Crystal Formation.

Risk Event – TR3	Crystal cracking or No Crystal Formation	
Probability	L	From reproducible ground testing, the probability of crystal cracking or no crystal formation under standard gravity is very low. The physical processes driving initial crystal formation are independent of gravity.
Severity	H	If crystals become dry before analysis and crack or do not form, there will be no sample to analyse.
Mitigation Plan	Performing ground testing of ten sample before the flight will allow for optimization of collection to prevent crystal cracking and optimise crystal formation.	
Contingency Plan	If crystal formation occurs in ground conditions but not in microgravity conditions, this points to the need for further investigation as it may point to unconsidered factors in the crystal formation process.	

Table 28: Electrical system failure under microgravity or high acceleration.

Risk Event – TR4	Electrical system failure under microgravity or high acceleration.	
Probability	L	There is a possibility of a plug or electrical connection come lose during flight.
Severity	H	Loss of power to a critical component during flight will end the experiment.
Mitigation Plan	Plugs and electrical connections will be secured with secondary adhesive such as hot glue to their respective connection points to prevent disconnection under high loads.	
Contingency Plan	If during the first flight electrical power is lost to a critical component (translation stage, laser, or controller) the experiment will be ceased and the power to box will be shutdown. The cause of electrical failure will be analysis between flights and if the cause is identified the problem will be rectified on site. If unable to identify why the problem occurred a replacement component will be put in for the connection.	

Table 29: Camera failure or poor image acquisition.

Risk Event – TR5	Camera failure or poor image acquisition	
Probability	L	Low probability of the camera failing in flight or having change in image quality during flight if ground testing on vibration table showed reasonable image quality.
Severity	L	Imaging is not required for the experiment.
Mitigation Plan	No mitigation.	
Contingency Plan	Continue experiment without camera.	

Table 30: Data corruption or loss during flight.

Risk Event – TR6	Data corruption or loss during flight	
Probability	L	Low probability of the Raspberry Pi corrupting data during flight.
Severity	M	Loss of supporting information for our findings.
Mitigation Plan	Data will be backed up to the controlling laser between trials so that only one datapoint would be lost if the data on the Raspberry pi was to become corrupted or lost during flight.	
Contingency Plan	The Raspberry Pi assembly would be replaced between flights following data corruption.	

Appendix B: Laser Safety Calculations

B 1.0 Laser Specifications

Manufacturer: Glucoloop AG, Zurich Switzerland

Product name: IVY

Serial Number: 0009

Table 31: Manufacturer Performance Specifications from the manufacturer data sheet and operations manual.

Manufacturer Performance Specifications (29.6 C, 29.4% RH, dry air purged)			
Wavelength	2723 nm	Pulse Duration	2.1 ns
Pulse Repetition Rate	1500 Hz	Horizontal beam diameter	2.013 mm
Average power	31 mW	Vertical beam diameter	2.042 mm
Energy per pulse	20.7 μ J	Beam Quality, Horizontal	<1.2
Polarization	Vertical	Beam Quality, Vertical	<1.2
Half divergence angle	<1 mrad		

The unfocused beam would then have a fluence of $6.59 \times 10^{-4} \text{ J/cm}^2$ and power density of 0.988 W/cm^2 .

B 2.0 Laser Safety information

Table 32: Maximum permissible exposure values for between 2.6 μ m and 1000 μ m from ANSI Z136.1.

Single pulse Ocular MPE from ANSI Z136.1, Tables 5a (2.6-1000 μ m)	0.01 J/cm^2
Single pulse Skin MPE from ANSI Z136.1, Tables 7 (2.6-1000 μ m)	0.01 J/cm^2
Average Power MPE for Thermal and Photochemical Hazards (Rule 2, ANSI Z136.1 Section 8.2.3)	$MPE(t) = 0.56t^{0.25} \text{ J/cm}^2$ $MPE(20 \text{ s}) = 1.18 \text{ J/cm}^2$ $N = fT = (1500)(20) = 30000$ $MPE_{Rule 2} = \frac{1.18}{30000}$ $= 3.93 \times 10^{-5} \text{ J/cm}^2$
Multiple-Pulse MPE for Thermal Hazards (Rule 3, ANSI Z136.1 Section 8.2.3)	$MPE_F(t, f) = 0.56t^{0.25}(ft)^{-0.25} \text{ J/cm}^2$

	$MPE_F = 0.09 \frac{J}{cm^2}$ $MPE_{AVP} = MPE_F(t, f) * f$ $MPE_{AVP} = 135 W/cm^2$
--	--

Given the significant changes in laser absorption mechanisms within the 2.6–3.1 μm range for pulse durations in the direct impulse vibrational excitation (DIVE) regime (≤ 5 ns)^{21–25}, compared to longer wavelengths, it is important to also consider experimentally determined damage thresholds. Reported values indicate that the minimum damage threshold for short-pulse mid-infrared (MIR) lasers is approximately 0.2 J/cm²²².

The ocular MPE used in the NOHD calculations was determined in accordance with the ANSI Z136.1 multiple-pulse rules (Rule 2), which govern the permissible exposure for this laser operating condition.

B 3.0 Nominal Ocular Hazard Distance (NOHD)

Unaltered beam:

$$NOHD = \frac{1}{\phi} \left[\left(\frac{4\Phi}{\pi MPE} \right) - a^2 \right]^{\frac{1}{2}}$$

a is the beam in cm

Φ is the fluence in J/cm^2

ϕ Divergence angle in radians

Plugging in our known values:

$$NOHD = \frac{1}{0.002} \left[\left(\frac{4(20.7 \times 10^{-6} J)}{\pi \left(3.93 \times 10^{-5} \frac{J}{cm^2} \right)} \right) - (0.2 cm)^2 \right]^{\frac{1}{2}} = 343 cm$$

Focused using 15 mm off axis parabolic mirror:

$$NOHD = \frac{f}{b} \left(\frac{4\Phi}{\pi MPE} \right)^{\frac{1}{2}}$$

b is the diameter of the incident beam on the focusing lens

f is the focal length of the focusing element.

Plugging in our known values:

$$NOHD = \frac{0.15}{0.2} \left(\frac{4(20.7 \times 10^{-6} J)}{\pi \left(3.93 \times 10^{-5} \frac{J}{cm^2} \right)} \right)^{\frac{1}{2}} = 0.61 cm$$

B 4.0 Nominal Hazard Zone (NHZ)

Diffuse reflection: $NHZ = \left(\frac{p_{\lambda} \Phi \cos(\theta)}{\pi MPE} \right)^{\frac{1}{2}}$

Assuming maximum reflection from the surface: $NHZ = \left(\frac{(1)(20.7 \times 10^{-6} \text{ J})(1)}{\pi(3.93 \times 10^{-5} \text{ J/cm}^2)} \right)^{\frac{1}{2}} = 0.41 \text{ cm}$

Specular reflection: $NHZ = \frac{1}{\phi} \left(\frac{4p_{\lambda} \Phi}{\pi MPE} \right)^{\frac{1}{2}}$

Assuming maximum reflection from the surface:

$$NHZ = \frac{1}{0.001} \left(\frac{(4)(1)(20.7 \times 10^{-6} \text{ J})(1)}{\pi(3.93 \times 10^{-5} \text{ J/cm}^2)} \right)^{\frac{1}{2}} = 819 \text{ cm} = 8.18 \text{ m}$$

If we consider the system with the focusing off axis parabolic mirror, $\phi = \frac{2\lambda}{\pi w_0} = 0.1119$

$$NHZ = \frac{1}{0.1119} \left(\frac{(4)(1)(20.7 \times 10^{-6} \text{ J})(1)}{\pi(3.93 \times 10^{-5} \text{ J/cm}^2)} \right)^{\frac{1}{2}} = 7.32 \text{ cm}$$

B 5.0 Optical Density (OD)

Unfocused beam OD requirement:

$$OD = \log \left(\frac{F}{MPE} \right) = \log \left(\frac{1.6 \times 10^{-4} \text{ J/cm}^2}{3.93 \times 10^{-5} \text{ J/cm}^2} \right) = 0.61$$

$$OD = \log \left(\frac{P_{Avg}}{MPE} \right) = \log \left(\frac{0.24 \text{ W/cm}^2}{135 \text{ W/cm}^2} \right) = -2.8$$

B 6.0 Conclusions

For the system used in the CRISTAR project, the NHZ and NOHD has the potential to be larger than the dimensions of the enclosure. As such, the system shall only be operated in a laser-safe laboratory with appropriate laser safety eyewear whenever the enclosure is open or the interlocks are intentionally defeated. During normal operation with the enclosure closed and interlocks engaged, the system functions as a Class 1 laser product. The laser safety eyewear used for servicing and alignment operations provides a minimum OD of 1 at the operating wavelength of 2.7 μm .

Appendix C: Flight Procedure Flowcharts

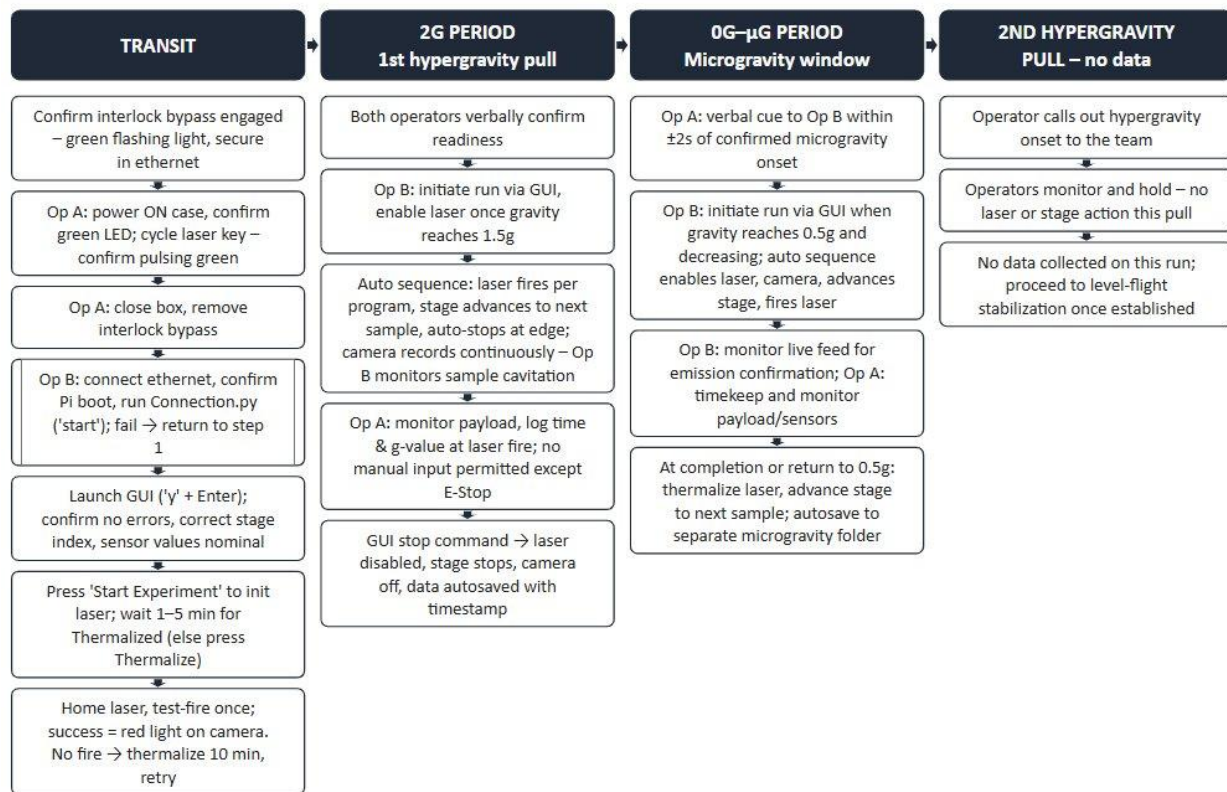


Figure C1: Procedural Flowcharts from Transit to 2nd Hyper gravity Pull

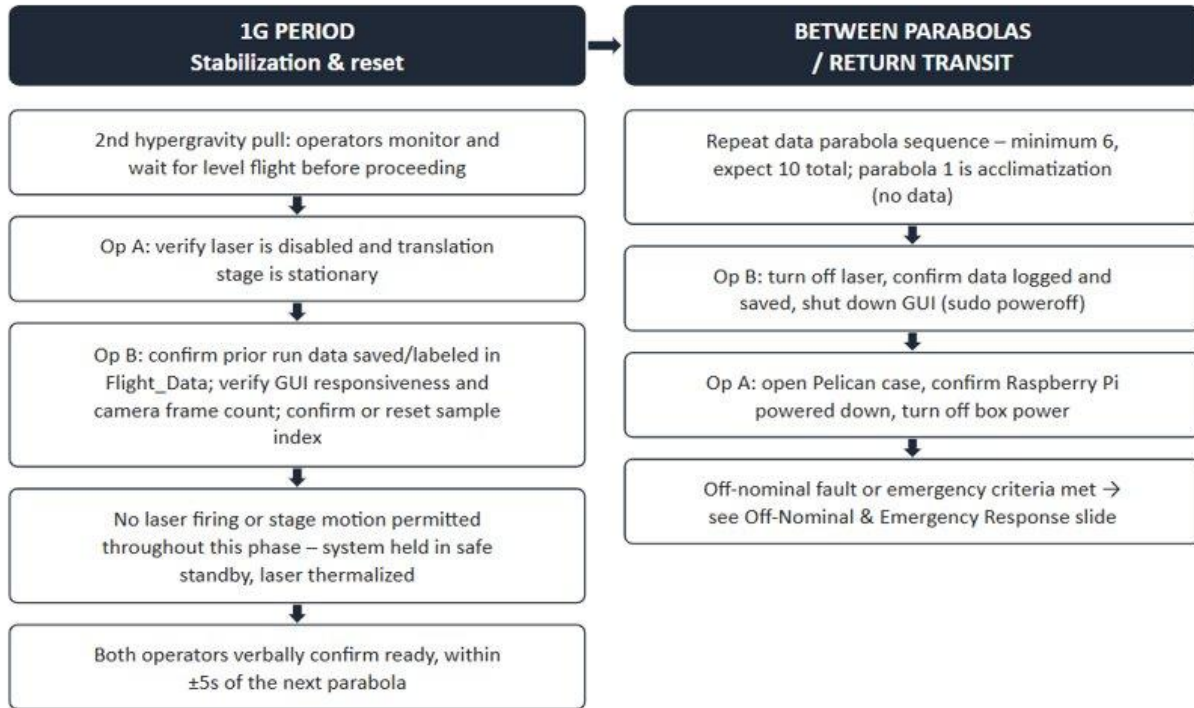


Figure C2: Procedural Flowchart for 1G Period, Between-Parabola, and Return Flight.

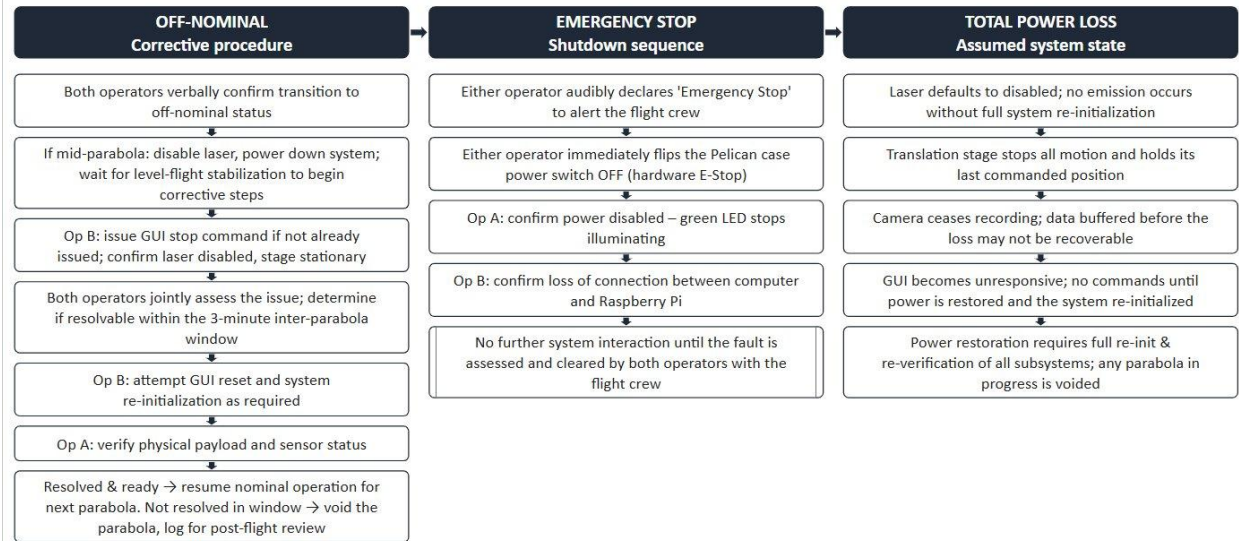


Figure C3: Procedural Flowchart for Off-Nominal, Emergency and Total Power Loss Events.

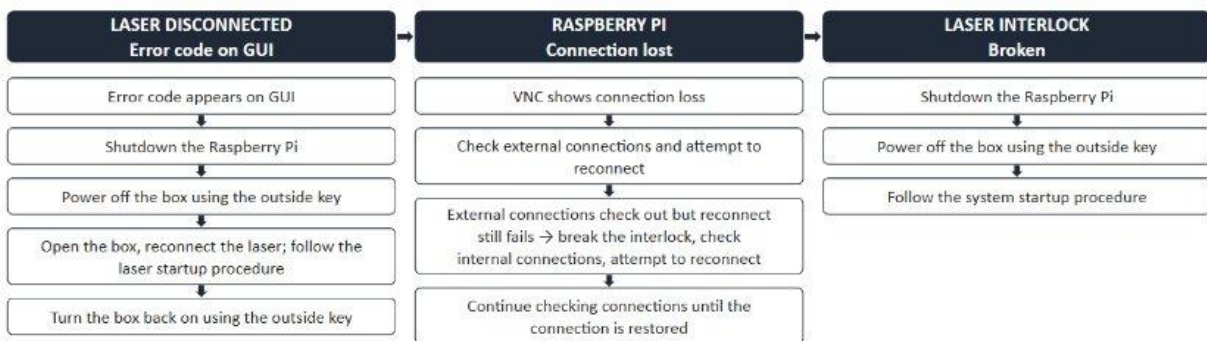


Figure C4: Procedural Flowchart for Component Specific Fault Recovery.

Appendix C: Safety Data Sheets (SDSs)

• SAFETY DATA SHEET

Version 6.11
Revision Date 28.09.2025
Print Date 12.08.2026

SECTION 1. IDENTIFICATION

1.1 Product identifiers

Product name : Glycine

Product Number : G7126
Brand : Sigma
CAS-No. : 56-40-6

1.2 Relevant identified uses of the substance or mixture and uses advised against

Identified uses : Laboratory chemicals, Synthesis of substances

1.3 Details of the supplier of the safety data sheet

Company : MilliporeSigma Canada Ltd.
2149 WINSTON PARK DRIVE
OAKVILLE ON L6H 6J8
CANADA

Telephone : +1 905 829-9500
Fax : +1 905 829-9292

1.4 Emergency telephone number

Emergency Phone # : +1-703-527-3887 CHEMTREC
(International)
24 Hours/day; 7 Days/week

SECTION 2. HAZARDS IDENTIFICATION

GHS classification in accordance with the Hazardous Products Regulations

Not a hazardous substance or mixture.

GHS label elements

No hazard pictogram, no signal word, no hazard statement(s), no precautionary statement(s) required.

Other hazards

None known.

SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

Substance / Mixture : Substance

Sigma - G7126

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Components

No hazardous ingredients

SECTION 4. FIRST AID MEASURES

If inhaled	: After inhalation: fresh air.
In case of skin contact	: In case of skin contact: Take off immediately all contaminated clothing. Rinse skin with water/ shower.
In case of eye contact	: After eye contact: rinse out with plenty of water. Remove contact lenses.
If swallowed	: After swallowing: make victim drink water (two glasses at most). Consult doctor if feeling unwell.
Most important symptoms and effects, both acute and delayed	: The most important known symptoms and effects are described in the labelling (see section 2.2) and/or in section 11
Protection of first-aiders	: For personal protection see section 8.
Notes to physician	: No data available

SECTION 5. FIREFIGHTING MEASURES

Suitable extinguishing media	: Water Foam Carbon dioxide (CO ₂) Dry powder
Unsuitable extinguishing media	: For this substance/mixture no limitations of extinguishing agents are given.
Specific hazards during fire fighting	: Combustible. Development of hazardous combustion gases or vapours possible in the event of fire.
Hazardous combustion products	: Carbon oxides Nitrogen oxides (NO _x)

Specific extinguishing methods	: No data available
Further information	: Suppress (knock down) gases/vapours/mists with a water spray jet. Prevent fire extinguishing water from contaminating surface water or the ground water system.
Special protective equipment for fire-fighters	: In the event of fire, wear self-contained breathing apparatus.

SECTION 6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures	: Advice for non-emergency personnel: Avoid inhalation of dusts. Evacuate the danger area, observe emergency procedures, consult an expert. Advice for emergency responders: For personal protection see section 8.
Environmental precautions	: Do not let product enter drains.
Methods and materials for containment and cleaning up	: Cover drains. Collect, bind, and pump off spills. Observe possible material restrictions (see sections 7 and 10). Take up dry. Dispose of properly. Clean up affected area. Avoid generation of dusts.

SECTION 7. HANDLING AND STORAGE

For precautions see section 2.2.

Further information on storage conditions	: Tightly closed. Dry.
Storage class	: 11, Combustible Solids
Recommended storage temperature	: Recommended storage temperature see product label.

SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Ingredients with workplace control parameters

Contains no substances with occupational exposure limit values.

Engineering measures : No data available

Personal protective equipment

Respiratory protection : required when dusts are generated.

Our recommendations on filtering respiratory protection are based on the following standards: DIN EN 143, DIN 14387 and other accompanying standards relating to the used respiratory protection system.

Recommended Filter type: : Filter type P1

The entrepreneur has to ensure that maintenance, cleaning and testing of respiratory protective devices are carried out according to the instructions of the producer. These measures have to be properly documented.

Hand protection

Material : Nitrile rubber
Break through time : 480 min
Glove thickness : 0.11 mm
Protective index : Full contact
Manufacturer : KCL 741 Dermatril® L

Material : Nitrile rubber
Break through time : 480 min
Glove thickness : 0.11 mm
Protective index : Splash contact
Manufacturer : KCL 741 Dermatril® L

Remarks : This recommendation applies only to the product stated in the safety data sheet, supplied by us and for the designated use. When dissolving in or mixing with other substances and under conditions deviating from those stated in EN 16523-1 please contact the supplier of CE-approved gloves (e.g. KCL GmbH, D-36124 Eichenzell, Internet: www.kcl.de).

Eye protection : Use equipment for eye protection tested and approved under appropriate government standards such as NIOSH (US) or EN 166(EU).
Safety glasses

Hygiene measures : Change contaminated clothing. Wash hands after working with substance.

SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance	: Crystalline powder
Color	: white
Odor	: odourless
Odor Threshold	: Not applicable
pH	: 5.9 - 6.4 (20 °C) Concentration: 50 g/l
Melting point/ range	: 240 °C
	: Not applicable
Flash point	: No data available
Evaporation rate	: No data available
Burning rate	: No data available
Self-ignition	: > 140 °C GLP: yes not auto-flammable
Upper explosion limit / Upper flammability limit	: No data available
Lower explosion limit / Lower flammability limit	: No data available
Vapor pressure	: No data available
Relative vapour density	: No data available
Relative density	: No data available
Density	: 1.161 g/cm ³ (20 °C)
Solubility(ies) Water solubility	: 250 g/l soluble (25 °C)
Partition coefficient: n- octanol/water	: log Pow: -3.21 Bioaccumulation is not expected. (ECHA)

Autoignition temperature	: No data available
Decomposition temperature	: > 233 °C
Viscosity, dynamic	: No data available
Viscosity, kinematic	: No data available
Flow time	: No data available
Explosive properties	: Not classified as explosive.
Oxidizing properties	: none
Molecular weight	: 75.07 g/mol
Particle characteristics Particle size	: No data available

SECTION 10. STABILITY AND REACTIVITY

Reactivity	: The following applies in general to flammable organic substances and mixtures: in correspondingly fine distribution, when whirled up a dust explosion potential may generally be assumed.
Chemical stability	: The product is chemically stable under standard ambient conditions (room temperature) .
Possibility of hazardous reactions	: Violent reactions possible with: Strong oxidizing agents Bases
Conditions to avoid	: no information available
Incompatible materials	: No data available
Hazardous decomposition products	: In the event of fire: see section 5

SECTION 11. TOXICOLOGICAL INFORMATION

11.1 Information on toxicological effects

Acute toxicity

LD50 Oral - Rat - 7,930 mg/kg

Remarks: (RTECS)

Inhalation: No data available

Dermal: No data available

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Skin corrosion/irritation

Skin - Rabbit

Result: No skin irritation - 4 h

(OECD Test Guideline 404)

Serious eye damage/eye irritation

Eyes - Rabbit

Result: No eye irritation - 72 h

(OECD Test Guideline 405)

Respiratory or skin sensitization

Local lymph node assay (LLNA) - Mouse

Result: negative

(OECD Test Guideline 429)

Germ cell mutagenicity

Test Type: In vitro mammalian cell gene mutation test

Test system: Mouse lymphoma test

Metabolic activation: with and without metabolic activation

Method: OECD Test Guideline 476

Result: negative

Test Type: Ames test

Test system: Escherichia coli/Salmonella typhimurium

Metabolic activation: with and without metabolic activation

Method: OECD Test Guideline 471

Result: negative

Test Type: Chromosome aberration test in vitro

Test system: Chinese hamster fibroblasts

Metabolic activation: without metabolic activation

Result: negative

Remarks: (ECHA)

Carcinogenicity

No data available

Reproductive toxicity

No data available

Specific target organ toxicity - single exposure

No data available

Specific target organ toxicity - repeated exposure

No data available

Aspiration hazard

No data available

11.2 Additional Information

RTECS: MB7600000

To the best of our knowledge, the chemical, physical, and toxicological properties have not been thoroughly investigated.

SECTION 12. ECOLOGICAL INFORMATION

Ecotoxicity

No data available

Persistence and degradability

No data available

Bioaccumulative potential

No data available

Mobility in soil

No data available

Other adverse effects

No data available

SECTION 13. DISPOSAL CONSIDERATIONS

Disposal methods

Waste from residues : Waste material must be disposed of in accordance with the national and local regulations. Leave chemicals in original containers. No mixing with other waste. Handle uncleaned containers like the product itself.

SECTION 14. TRANSPORT INFORMATION

International Regulations

IATA-DGR

Not regulated as a dangerous good

IMDG-Code

Not regulated as a dangerous good

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable for product as supplied.

National Regulations

TDG

Not regulated as a dangerous good

Special precautions for user

Remarks : Not classified as dangerous in the meaning of transport regulations.

SECTION 15. REGULATORY INFORMATION

Canadian lists

No substances are subject to a Significant New Activity Notification.

SECTION 16. OTHER INFORMATION

Full text of other abbreviations

AIIC - Australian Inventory of Industrial Chemicals; ANTT - National Agency for Transport by Land of Brazil; ASTM - American Society for the Testing of Materials; bw - Body weight; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; ERG - Emergency Response Guide; GHS - Globally Harmonised System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organisation; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardisation; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; MERCOSUR - The Agreement for the Facilitation of the Transport of Dangerous Goods; n.o.s. - Not Otherwise Specified; Nch - Chilean Norm; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NOM - Official Mexican Norm; NTP - National Toxicology Program; NZIoC - New Zealand Inventory of Chemicals; OECD - Organisation for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; TCSI - Taiwan Chemical Substance Inventory; TDG - Transportation of Dangerous Goods; TECI - Thailand Existing Chemicals Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative; WHMIS - Workplace Hazardous Materials Information System

The information is believed to be correct but is not exhaustive and will be used solely as a guideline, which is based on current knowledge of the chemical substance or mixture and is applicable to appropriate safety precautions for the product. It does not represent any guarantee of the properties of the product. Sigma-Aldrich Corporation and its Affiliates shall not be held liable for any damage resulting from handling or from contact with the above product. See www.sigma-aldrich.com and/or the reverse side of invoice or packing slip for additional terms and conditions of sale.

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CA / EN

• SAFETY DATA SHEET

Version 6.10
Revision Date 13.09.2025
Print Date 14.09.2025

SECTION 1. IDENTIFICATION

1.1 Product identifiers

Product name : Acetic acid

Product Number : A6283
Brand : SIGALD
Index-No. : 607-002-00-6
CAS-No. : 64-19-7

1.2 Relevant identified uses of the substance or mixture and uses advised against

Identified uses : Laboratory chemicals, Synthesis of substances

1.3 Details of the supplier of the safety data sheet

Company : MilliporeSigma Canada Ltd.
2149 WINSTON PARK DRIVE
OAKVILLE ON L6H 6J8
CANADA

Telephone : +1 905 829-9500
Fax : +1 905 829-9292

1.4 Emergency telephone number

Emergency Phone # : +1-703-527-3887 CHEMTREC
(International)
24 Hours/day; 7 Days/week

SECTION 2. HAZARDS IDENTIFICATION

GHS classification in accordance with the Hazardous Products Regulations

Flammable liquids : Category 3
Skin corrosion : Category 1A
Serious eye damage : Category 1

GHS label elements

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Hazard pictograms



Signal Word

: Danger

Hazard Statements

: H226 Flammable liquid and vapour.
H314 Causes severe skin burns and eye damage.

Precautionary statements :

Prevention:

P210 Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
P233 Keep container tightly closed.
P240 Ground and bond container and receiving equipment.
P241 Use explosion-proof electrical/ ventilating/ lighting equipment.
P242 Use non-sparking tools.
P243 Take action to prevent static discharges.
P264 Wash skin thoroughly after handling.
P280 Wear protective gloves/ protective clothing/ eye protection/ face protection.

Response:

P301 + P330 + P331 IF SWALLOWED: Rinse mouth. Do NOT induce vomiting.
P303 + P361 + P353 IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water.
P304 + P340 + P310 IF INHALED: Remove person to fresh air and keep comfortable for breathing. Immediately call a POISON CENTER/ doctor.
P305 + P351 + P338 + P310 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER/ doctor.
P363 Wash contaminated clothing before reuse.
P370 + P378 In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish.

Storage:

P403 + P235 Store in a well-ventilated place. Keep cool.
P405 Store locked up.

Disposal:

P501 Dispose of contents/ container to an approved waste disposal plant.

Other hazards

Lachrymator.

SIGALD - A6283

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SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

Substance / Mixture : Substance

Components

Chemical name	CAS-No.	Concentration (% w/w)
acetic acid	64-19-7	>= 80 - <= 100 *

* Actual concentration or concentration range is withheld as a trade secret

SECTION 4. FIRST AID MEASURES

General advice : First aiders need to protect themselves.
Show this safety data sheet to the doctor in attendance.

If inhaled : After inhalation: fresh air. Call in physician.

In case of skin contact : In case of skin contact: Take off immediately all contaminated clothing. Rinse skin with water/ shower.
Call a physician immediately.

In case of eye contact : After eye contact: rinse out with plenty of water.
Immediately call in ophthalmologist.
Remove contact lenses.

If swallowed : After swallowing: make victim drink water (two glasses at most), avoid vomiting (risk of perforation).
Call a physician immediately.
Do not attempt to neutralise.

Most important symptoms and effects, both acute and delayed : The most important known symptoms and effects are described in the labelling (see section 2.2) and/or in section 11

Protection of first-aiders : For personal protection see section 8.

Notes to physician : No data available

SECTION 5. FIREFIGHTING MEASURES

Suitable extinguishing media : Water
Foam
Carbon dioxide (CO₂)
Dry powder

Unsuitable extinguishing media : For this substance/mixture no limitations of extinguishing agents are given.

Specific hazards during fire fighting	: Combustible.
	Vapours are heavier than air and may spread along floors.
	Forms explosive mixtures with air at elevated temperatures.
	Development of hazardous combustion gases or vapours possible in the event of fire.
Hazardous combustion products	: Carbon oxides
Specific extinguishing methods	: No data available
Further information	: Remove container from danger zone and cool with water. Prevent fire extinguishing water from contaminating surface water or the ground water system.
Special protective equipment for fire-fighters	: Stay in danger area only with self-contained breathing apparatus. Prevent skin contact by keeping a safe distance or by wearing suitable protective clothing.

SECTION 6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures	: Advice for non-emergency personnel: Do not breathe vapours, aerosols. Avoid substance contact. Ensure adequate ventilation. Keep away from heat and sources of ignition. Evacuate the danger area, observe emergency procedures, consult an expert. Advice for emergency responders: For personal protection see section 8.
Environmental precautions	: Do not let product enter drains. Risk of explosion.
Methods and materials for containment and	: Cover drains. Collect, bind, and pump off spills. Observe possible material restrictions (see sections 7

cleaning up

and 10).

Take up with liquid-absorbent and neutralising material (e.g. Chemizorb® H⁺, Merck Art. No. 101595). Dispose of properly. Clean up affected area.

SECTION 7. HANDLING AND STORAGE

For precautions see section 2.2.

Advice on protection against fire and explosion	: Keep away from open flames, hot surfaces and sources of ignition. Take precautionary measures against static discharge.
Further information on storage conditions	: Keep container tightly closed in a dry and well-ventilated place. Keep away from heat and sources of ignition.
Storage class	: 3, Flammable liquids
Recommended storage temperature	: Recommended storage temperature see product label.
Further information on storage stability	: Moisture sensitive.
Packaging material	: Suitable material: Poly Drum

SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Ingredients with workplace control parameters

Components	CAS-No.	Value type (Form of exposure)	Control parameters / Permissible concentration	Basis
acetic acid	64-19-7	TWA	10 ppm 25 mg/m ³	CA AB OEL
		STEL	15 ppm 37 mg/m ³	CA AB OEL
		TWA	10 ppm	CA BC OEL
		STEL	15 ppm	CA BC OEL
		TWAEV	10 ppm 25 mg/m ³	CA QC OEL
		STEV	15 ppm 37 mg/m ³	CA QC OEL
		TWA	10 ppm	ACGIH
		STEL	15 ppm	ACGIH

Engineering measures : No data available

Personal protective equipment

Respiratory protection : required when vapours/aerosols are generated.
Our recommendations on filtering respiratory protection are based on the following standards: DIN EN 143, DIN 14387 and other accompanying standards relating to the used respiratory protection system.

Recommended Filter type: : filter E-(P2)

The entrepreneur has to ensure that maintenance, cleaning and testing of respiratory protective devices are carried out according to the instructions of the producer. These measures have to be properly documented.

Hand protection

Material : butyl-rubber
Break through time : 480 min
Glove thickness : 0.7 mm
Protective index : Full contact
Manufacturer : Butoject® (KCL 898)

Material : Latex gloves
Break through time : 30 min
Glove thickness : 0.6 mm
Protective index : Splash contact
Manufacturer : Lapren® (KCL 706 / Aldrich Z677558, Size M)

Remarks : This recommendation applies only to the product stated in the safety data sheet, supplied by us and for the designated use. When dissolving in or mixing with other substances and under conditions deviating from those stated in EN 16523-1 please contact the supplier of CE-approved gloves (e.g. KCL GmbH, D-36124 Eichenzell, Internet: www.kcl.de).

Eye protection : Use equipment for eye protection tested and approved under appropriate government standards such as NIOSH (US) or EN 166(EU).
Tightly fitting safety goggles

Skin and body protection : Flame retardant antistatic protective clothing.

Hygiene measures : Immediately change contaminated clothing. Apply preventive skin protection. Wash hands and face after working with substance.

SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance	: liquid
Color	: colourless
Odor	: stinging
Odor Threshold	: 0.2 ppm
pH	: 2.5 (20 °C) Concentration: 50 g/l
Melting point/ range	: 16.2 °C Method: lit.
Boiling point/boiling range	: 117 - 118 °C Method: lit.
Flash point	: 39 °C (1,013 hPa) Method: closed cup, closed cup
Evaporation rate	: No data available
Flammability (solid, gas)	: No data available
Flammability (liquids)	: No data available
Burning rate	: No data available
Self-ignition	: 463 °C
Upper explosion limit / Upper flammability limit	: Upper explosion limit 19.9 %(V) (59.0 °C)
Lower explosion limit / Lower flammability limit	: Lower explosion limit 4 %(V)
Vapor pressure	: 20.79 hPa (25 °C)
Relative vapour density	: 2.07
Relative density	: No data available
Density	: 1.049 g/cm ³ (25 °C) Method: lit.

Solubility(ies)
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Water solubility	: 602.9 g/l completely soluble (25 °C, 1,013 hPa) pH: 7
Partition coefficient: n-octanol/water	: log Pow: -0.17 (25 °C) pH: 7 Method: (experimental) Bioaccumulation is not expected. (ECHA)
Autoignition temperature	: 463 °C
Decomposition temperature	: Distillable in an undecomposed state at normal pressure.
Viscosity	
Viscosity, dynamic	: 1.05 mPa.s (25 °C)
Viscosity, kinematic	: 1.17 mm ² /s (20 °C)
Flow time	: No data available
Explosive properties	: Not classified as explosive.
Oxidizing properties	: none
Surface tension	: 28.8 mN/m, 10.0 °C
Refractive index	: 1.37 (20 °C)
Molecular weight	: 60.05 g/mol
Particle characteristics	
Particle size	: No data available

SECTION 10. STABILITY AND REACTIVITY

Reactivity	: Vapour/air-mixtures are explosive at intense warming.
Chemical stability	: The product is chemically stable under standard ambient conditions (room temperature) .
Possibility of hazardous reactions	: Risk of explosion with: peroxi compounds perchloric acid fuming sulfuric acid phosphorus halides hydrogen peroxide chromium(VI) oxide potassium permanganate Peroxides

Strong oxidizing agents
Risk of ignition or formation of inflammable gases or vapours with:
Iron
Zinc
magnesium
Mild steel
Possible formation of:
Hydrogen
Violent reactions possible with:
strong alkalis
Aldehydes
alkali hydroxides
nonmetallic halides
ethanolamine
Acetaldehyde
Alcohols
halogen-halogen compounds
chlorosulfonic acid
chromosulfuric acid
Potassium hydroxide
Nitric acid

Conditions to avoid : Heating.

Incompatible materials : No data available

Hazardous decomposition products : In the event of fire: see section 5

SECTION 11. TOXICOLOGICAL INFORMATION

11.1 Information on toxicological effects

Acute toxicity

LD50 Oral - Rat - 3,310 mg/kg

Remarks: (RTECS)

LC50 Inhalation - Mouse - 4 h - 2,819 mg/l - vapour

Remarks: (RTECS)

Dermal: No data available

Skin corrosion/irritation

Skin - Rabbit

Result: Causes burns. - 4 h

(OECD Test Guideline 404)

Remarks: Classified according to Regulation (EU) 1272/2008, Annex VI (Table 3.1/3.2)

Serious eye damage/eye irritation

Eyes - Rabbit

Result: Causes burns. - 4 h

(OECD Test Guideline 405)

Remarks: (IUCLID)

Remarks: Causes serious eye damage.

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Respiratory or skin sensitization

No data available

Germ cell mutagenicity

Test Type: Ames test

Test system: Salmonella typhimurium

Metabolic activation: with and without metabolic activation

Method: OECD Test Guideline 471

Result: negative

Test Type: Mutagenicity (mammal cell test): chromosome aberration.

Test system: Chinese hamster ovary cells

Metabolic activation: with and without metabolic activation

Method: OECD Test Guideline 473

Result: negative

Test Type: Micronucleus test

Species: Rat

Cell type: Bone marrow

Application Route: inhalation (vapour)

Method: Mutagenicity (micronucleus test)

Result: negative

Carcinogenicity

No data available

Reproductive toxicity

No data available

Specific target organ toxicity - single exposure

No data available

Specific target organ toxicity - repeated exposure

No data available

Aspiration hazard

No data available

11.2 Additional Information

RTECS: AF1225000

Material is extremely destructive to tissue of the mucous membranes and upper respiratory tract, eyes, and skin., spasm, inflammation and edema of the larynx, spasm, inflammation and edema of the bronchi, pneumonitis, pulmonary edema, burning sensation, Cough, wheezing, laryngitis, Shortness of breath, Headache, Nausea, Vomiting, Ingestion or inhalation of concentrated acetic acid causes damage to tissues of the respiratory and digestive tracts. Symptoms include: hematemesis, bloody diarrhea, edema and/or perforation of the esophagus and pylorus, pancreatitis, hematuria, anuria, uremia, albuminuria, hemolysis, convulsions, bronchitis, pulmonary edema, pneumonia, cardiovascular collapse, shock, and death. Direct contact or exposure to high concentrations of vapor with skin or eyes can cause: erythema, blisters, tissue destruction with slow healing, skin blackening, hyperkeratosis, fissures, corneal erosion, opacification, iritis, conjunctivitis, and possible blindness.

To the best of our knowledge, the chemical, physical, and toxicological properties have not been thoroughly investigated.

Stomach - Irregularities - Based on Human Evidence

SECTION 12. ECOLOGICAL INFORMATION

Ecotoxicity

Components:

acetic acid:

Toxicity to fish	: LC50 (Oncorhynchus mykiss (rainbow trout)): > 1,000 mg/l End point: mortality Exposure time: 96 h Test Type: semi-static test Method: OECD Test Guideline 203 GLP: yes
Toxicity to daphnia and other aquatic invertebrates	: EC50 (Daphnia magna (Water flea)): > 1,000 mg/l End point: Immobilization Exposure time: 48 h Test Type: static test Analytical monitoring: yes Method: OECD Test Guideline 202 GLP: yes
Toxicity to algae/aquatic plants	: EC50 (Skeletonema costatum): > 1,000 mg/l Exposure time: 72 h Test Type: static test Method: ISO 10253 GLP: yes
Toxicity to microorganisms	: EC5 (Pseudomonas putida): 2,850 mg/l Exposure time: 16 h Remarks: neutral (maximum permissible toxic concentration) (Lit.) EC50 (Photobacterium phosphoreum): 11 mg/l Exposure time: 15 min Test Type: microtox test Remarks: (IUCLID)

Persistence and degradability

Components:

acetic acid:

Biodegradability	: Result: Readily biodegradable. Biodegradation: 99 % Exposure time: 30 d Method: OECD Test Guideline 301D
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Remarks: (HSDB)

Result: Readily eliminated from water
Biodegradation: 95 %
Exposure time: 5 d
Method: OECD Test Guideline 302B

Biochemical Oxygen Demand (BOD) : 880 mg/g
Incubation time: 5 d
Remarks: (Lit.)

BOD/ThOD : 76 %
Remarks: (IUCLID)

Bioaccumulative potential

Components:

acetic acid:

Partition coefficient: n-octanol/water : log Pow: -0.17 (25 °C)
pH: 7
Method: (experimental)
Remarks: Bioaccumulation is not expected.
(ECHA)

Mobility in soil

No data available

Other adverse effects

Components:

acetic acid:

Results of PBT and vPvB assessment : Substance does not meet the criteria for PBT or vPvB according to Regulation (EC) No 1907/2006, Annex XIII.

SECTION 13. DISPOSAL CONSIDERATIONS

Disposal methods

Waste from residues : Waste material must be disposed of in accordance with the national and local regulations. Leave chemicals in original containers. No mixing with other waste. Handle uncleaned containers like the product itself.

SECTION 14. TRANSPORT INFORMATION

International Regulations

IATA-DGR

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UN/ID No. : UN 2789
Proper shipping name : Acetic acid, glacial
Class : 8
Subsidiary risk : 3
Packing group : II
Labels : Class 8 - Corrosive substances, Class 3 - Flammable liquids
Packing instruction (cargo : 855
aircraft)
Packing instruction : 851
(passenger aircraft)

IMDG-Code

UN number : UN 2789
Proper shipping name : ACETIC ACID, GLACIAL

Class : 8
Subsidiary risk : 3
Packing group : II
Labels : 8 (3)
EmS Code : F-E, S-C
Marine pollutant : no

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable for product as supplied.

National Regulations

TDG

UN number : UN 2789
Proper shipping name : ACETIC ACID, GLACIAL

Class : 8
Subsidiary risk : 3
Packing group : II
Labels : 8 (3)
ERG Code : 132
Marine pollutant : no

Special precautions for user

The transport classification(s) provided herein are for informational purposes only, and solely based upon the properties of the unpackaged material as it is described within this Safety Data Sheet. Transportation classifications may vary by mode of transportation, package sizes, and variations in regional or country regulations.

SECTION 15. REGULATORY INFORMATION

Canadian lists

No substances are subject to a Significant New Activity Notification.

SECTION 16. OTHER INFORMATION

Full text of other abbreviations

ACGIH	: USA. ACGIH Threshold Limit Values (TLV)
CA AB OEL	: Canada. Alberta, Occupational Health and Safety Code (table 2: OEL)
CA BC OEL	: Canada. British Columbia OEL
CA QC OEL	: Québec. Regulation respecting occupational health and safety, Schedule 1, Part 1: Permissible exposure values for airborne contaminants
ACGIH / TWA	: 8-hour, time-weighted average
ACGIH / STEL	: Short-term exposure limit
CA AB OEL / TWA	: 8-hour Occupational exposure limit
CA AB OEL / STEL	: 15-minute occupational exposure limit
CA BC OEL / TWA	: 8-hour time weighted average
CA BC OEL / STEL	: short-term exposure limit
CA QC OEL / TWAEV	: Time-weighted average exposure value
CA QC OEL / STEV	: Short-term exposure value

AIIC - Australian Inventory of Industrial Chemicals; ANTT - National Agency for Transport by Land of Brazil; ASTM - American Society for the Testing of Materials; bw - Body weight; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; ERG - Emergency Response Guide; GHS - Globally Harmonised System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organisation; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardisation; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; MERCOSUR - The Agreement for the Facilitation of the Transport of Dangerous Goods; n.o.s. - Not Otherwise Specified; Nch - Chilean Norm; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NOM - Official Mexican Norm; NTP - National Toxicology Program; NZIoC - New Zealand Inventory of Chemicals; OECD - Organisation for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; TCSI - Taiwan Chemical Substance Inventory; TDG - Transportation of Dangerous Goods; TECI - Thailand Existing Chemicals Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB -

Very Persistent and Very Bioaccumulative; WHMIS - Workplace Hazardous Materials Information System

The information is believed to be correct but is not exhaustive and will be used solely as a guideline, which is based on current knowledge of the chemical substance or mixture and is applicable to appropriate safety precautions for the product. It does not represent any guarantee of the properties of the product. Sigma-Aldrich Corporation and its Affiliates shall not be held liable for any damage resulting from handling or from contact with the above product. See www.sigma-aldrich.com and/or the reverse side of invoice or packing slip for additional terms and conditions of sale.

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CA / EN

• SAFETY DATA SHEET

Version 6.9
Revision Date 07.11.2025
Print Date 06.03.2026

SECTION 1. IDENTIFICATION

1.1 Product identifiers

Product name : Lysozyme, from chicken egg white

Product Number : L6876
Brand : Sigma
CAS-No. : 12650-88-3

1.2 Relevant identified uses of the substance or mixture and uses advised against

Identified uses : Laboratory chemicals, Synthesis of substances

1.3 Details of the supplier of the safety data sheet

Company : MilliporeSigma Canada Ltd.
2149 WINSTON PARK DRIVE
OAKVILLE ON L6H 6J8
CANADA

Telephone : +1 905 829-9500
Fax : +1 905 829-9292

1.4 Emergency telephone number

Emergency Phone # : +1-703-527-3887 CHEMTREC
(International)
24 Hours/day; 7 Days/week

SECTION 2. HAZARDS IDENTIFICATION

GHS classification in accordance with the Hazardous Products Regulations

Respiratory sensitisation : Category 1

GHS label elements

Hazard pictograms :



Signal word : Danger

Hazard statements : H334 May cause allergy or asthma symptoms or

breathing difficulties if inhaled.

Precautionary statements :

Prevention:

P261 Avoid breathing dust.

P284 Wear respiratory protection.

Response:

P304 + P340 IF INHALED: Remove person to fresh air and keep comfortable for breathing.

P342 + P311 If experiencing respiratory symptoms: Call a POISON CENTER/ doctor.

Disposal:

P501 Dispose of contents/ container to an approved waste disposal plant.

Other hazards

None known.

SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

Substance / Mixture : Substance

CAS-No. : 12650-88-3

Components

Chemical name	CAS-No.	Concentration (% w/w)
Lysozyme	12650-88-3	>= 80 - <= 100 *

* Actual concentration or concentration range is withheld as a trade secret

SECTION 4. FIRST AID MEASURES

General advice : First aiders need to protect themselves.
Show this safety data sheet to the doctor in attendance.

If inhaled : After inhalation: fresh air. Call in physician.

In case of skin contact : In case of skin contact: Take off immediately all contaminated clothing. Rinse skin with water/ shower.

In case of eye contact : After eye contact: rinse out with plenty of water.
Remove contact lenses.

If swallowed : After swallowing: immediately make victim drink water (two glasses at most).
Consult a physician.

Most important : The most important known symptoms and effects are

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symptoms and effects, both acute and delayed	described in the labelling (see section 2.2) and/or in section 11
Protection of first-aiders	: For personal protection see section 8.
Notes to physician	: No data available

SECTION 5. FIREFIGHTING MEASURES

Suitable extinguishing media	: Water Foam Carbon dioxide (CO ₂) Dry powder
Unsuitable extinguishing media	: For this substance/mixture no limitations of extinguishing agents are given.
Specific hazards during fire fighting	: Combustible. Development of hazardous combustion gases or vapours possible in the event of fire.
Hazardous combustion products	: Nature of decomposition products not known.
Specific extinguishing methods	: No data available
Further information	: Prevent fire extinguishing water from contaminating surface water or the ground water system.
Special protective equipment for fire- fighters	: In the event of fire, wear self-contained breathing apparatus.

SECTION 6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures	: Advice for non-emergency personnel: Avoid inhalation of dusts. Avoid substance contact. Ensure adequate ventilation. Evacuate the danger area, observe emergency procedures, consult an expert.
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Advice for emergency responders:
For personal protection see section 8.

Environmental precautions	: Do not let product enter drains.
Methods and materials for containment and cleaning up	: Cover drains. Collect, bind, and pump off spills. Observe possible material restrictions (see sections 7 and 10). Take up carefully. Dispose of properly. Clean up affected area. Avoid generation of dusts.

SECTION 7. HANDLING AND STORAGE

For precautions see section 2.2.

Advice on safe handling	: Work under hood. Do not inhale substance/mixture.
Further information on storage conditions	: Tightly closed. Dry. Keep locked up or in an area accessible only to qualified or authorised persons.
Storage class	: 11, Combustible Solids
Recommended storage temperature	: 2 - 30 °C

SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Ingredients with workplace control parameters

Contains no substances with occupational exposure limit values.

Engineering measures : No data available

Personal protective equipment

Respiratory protection : required when dusts are generated.

Our recommendations on filtering respiratory protection are based on the following standards: DIN EN 143, DIN 14387 and other accompanying standards relating to the used respiratory protection system.

Recommended Filter type: : Filter type P2

The entrepreneur has to ensure that maintenance, cleaning and testing of respiratory protective devices are carried out according to the instructions of the producer. These measures have to be properly documented.

Hand protection

Material : Nitrile rubber
Break through time : 480 min
Glove thickness : 0.11 mm
Protective index : Full contact
Manufacturer : KCL 741 Dermatril® L

Material : Nitrile rubber
Break through time : 480 min
Glove thickness : 0.11 mm
Protective index : Splash contact
Manufacturer : KCL 741 Dermatril® L

Remarks : This recommendation applies only to the product stated in the safety data sheet, supplied by us and for the designated use. When dissolving in or mixing with other substances and under conditions deviating from those stated in EN 16523-1 please contact the supplier of CE-approved gloves (e.g. KCL GmbH, D-36124 Eichenzell, Internet: www.kcl.de).

Eye protection : Use equipment for eye protection tested and approved under appropriate government standards such as NIOSH (US) or EN 166(EU).
Safety glasses

Skin and body protection : protective clothing

Hygiene measures : Change contaminated clothing. Preventive skin protection recommended. Wash hands after working with substance.

SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance : powder, (lyophilised)

Color : beige

Odor : No data available

Odor Threshold : No data available

pH : No data available

Melting point	: No data available
Boiling point/boiling range	: No data available
Flash point	: No data available
Evaporation rate	: No data available
Flammability (solid, gas)	: No data available
Flammability (liquids)	: No data available
Burning rate	: No data available
Upper explosion limit / Upper flammability limit	: No data available
Lower explosion limit / Lower flammability limit	: No data available
Vapor pressure	: No data available
Relative vapour density	: No data available
Relative density	: No data available
Density	: No data available
Solubility(ies) Water solubility	: 5 g/l soluble in cold water
Partition coefficient: n- octanol/water	: No data available
Autoignition temperature	: No data available
Decomposition temperature	: No data available
Viscosity, dynamic	: No data available
Viscosity, kinematic	: No data available
Flow time	: No data available
Explosive properties	: Not classified as explosive.
Oxidizing properties	: none
Molecular weight	: 14,388 g/mol

Particle characteristics
Particle size : No data available

SECTION 10. STABILITY AND REACTIVITY

Reactivity : The following applies in general to flammable organic substances and mixtures: in correspondingly fine distribution, when whirled up a dust explosion potential may generally be assumed.

Chemical stability : The product is chemically stable under standard ambient conditions (room temperature) .

Possibility of hazardous reactions : Violent reactions possible with:
Strong oxidizing agents

Conditions to avoid : no information available

Incompatible materials : Strong oxidizing agents

Hazardous decomposition products : In the event of fire: see section 5

SECTION 11. TOXICOLOGICAL INFORMATION

11.1 Information on toxicological effects

Acute toxicity

LD50 Oral - Rat - male and female - > 5,050 mg/kg
(US-EPA)

LC50 Inhalation - Rat - male and female - 4 h - > 2.51 mg/l - dust/mist

(US-EPA)

LD50 Dermal - Rabbit - male and female - > 5,050 mg/kg
(US-EPA)

No data available

Skin corrosion/irritation

No data available

Serious eye damage/eye irritation

No data available

Respiratory or skin sensitization

Prolonged or repeated exposure may cause allergic reactions in certain sensitive individuals.

Germ cell mutagenicity

No data available

Carcinogenicity

No data available

Reproductive toxicity

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No data available

Specific target organ toxicity - single exposure

No data available

Specific target organ toxicity - repeated exposure

No data available

Aspiration hazard

No data available

11.2 Additional Information

RTECS: OL5989000

To the best of our knowledge, the chemical, physical, and toxicological properties have not been thoroughly investigated.

SECTION 12. ECOLOGICAL INFORMATION

Ecotoxicity

No data available

Persistence and degradability

No data available

Bioaccumulative potential

No data available

Mobility in soil

No data available

Other adverse effects

No data available

SECTION 13. DISPOSAL CONSIDERATIONS

Disposal methods

Waste from residues : Waste material must be disposed of in accordance with the national and local regulations. Leave chemicals in original containers. No mixing with other waste. Handle uncleaned containers like the product itself.

SECTION 14. TRANSPORT INFORMATION

International Regulations

IATA-DGR

Not regulated as a dangerous good

IMDG-Code

Not regulated as a dangerous good

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable for product as supplied.

National Regulations

TDG

Not regulated as a dangerous good

Special precautions for user

Remarks : Not classified as dangerous in the meaning of transport regulations.

SECTION 15. REGULATORY INFORMATION

Canadian lists

No substances are subject to a Significant New Activity Notification.

SECTION 16. OTHER INFORMATION

Full text of other abbreviations

AIIC - Australian Inventory of Industrial Chemicals; ANTT - National Agency for Transport by Land of Brazil; ASTM - American Society for the Testing of Materials; bw - Body weight; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; ERG - Emergency Response Guide; GHS - Globally Harmonised System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organisation; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardisation; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; MERCOSUR - The Agreement for the Facilitation of the Transport of Dangerous Goods; n.o.s. - Not Otherwise Specified; Nch - Chilean Norm; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NOM - Official Mexican Norm; NTP - National Toxicology Program; NZIoC - New Zealand Inventory of Chemicals; OECD - Organisation for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; TCSI - Taiwan Chemical Substance Inventory; TDG - Transportation of Dangerous Goods; TECI - Thailand Existing Chemicals

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Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative; WHMIS - Workplace Hazardous Materials Information System

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• SAFETY DATA SHEET

Version 7.3
Revision Date 10.09.2025
Print Date 11.09.2025

SECTION 1. IDENTIFICATION

1.1 Product identifiers

Product name : Sodium chloride

Product Number : S9888
Brand : SIGALD
CAS-No. : 7647-14-5

1.2 Relevant identified uses of the substance or mixture and uses advised against

Identified uses : Laboratory chemicals, Synthesis of substances

1.3 Details of the supplier of the safety data sheet

Company : MilliporeSigma Canada Ltd.
2149 WINSTON PARK DRIVE
OAKVILLE ON L6H 6J8
CANADA

Telephone : +1 905 829-9500
Fax : +1 905 829-9292

1.4 Emergency telephone number

Emergency Phone # : +1-703-527-3887 CHEMTREC
(International)
24 Hours/day; 7 Days/week

SECTION 2. HAZARDS IDENTIFICATION

GHS classification in accordance with the Hazardous Products Regulations

Not a hazardous substance or mixture.

GHS label elements

No hazard pictogram, no signal word, no hazard statement(s), no precautionary statement(s) required.

Other hazards

None known.

SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

Substance / Mixture : Substance

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Components

No hazardous ingredients

SECTION 4. FIRST AID MEASURES

If inhaled	: After inhalation: fresh air.
In case of skin contact	: In case of skin contact: Take off immediately all contaminated clothing. Rinse skin with water/ shower.
In case of eye contact	: After eye contact: rinse out with plenty of water. Remove contact lenses.
If swallowed	: After swallowing: make victim drink water (two glasses at most). Consult doctor if feeling unwell.
Most important symptoms and effects, both acute and delayed	: The most important known symptoms and effects are described in the labelling (see section 2.2) and/or in section 11
Protection of first-aiders	: For personal protection see section 8.
Notes to physician	: No data available

SECTION 5. FIREFIGHTING MEASURES

Suitable extinguishing media	: Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
Unsuitable extinguishing media	: For this substance/mixture no limitations of extinguishing agents are given.
Specific hazards during fire fighting	: Not combustible.

Ambient fire may liberate hazardous vapours.

Hazardous combustion products	: Hydrogen chloride gas
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Sodium oxides

Specific extinguishing methods	: No data available
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Further information	: Suppress (knock down) gases/vapours/mists with a water spray jet. Prevent fire extinguishing water from contaminating surface water or the ground water system.
Special protective equipment for fire-fighters	: In the event of fire, wear self-contained breathing apparatus.

SECTION 6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures	: Advice for non-emergency personnel: Avoid inhalation of dusts. Evacuate the danger area, observe emergency procedures, consult an expert. Advice for emergency responders: For personal protection see section 8.
Environmental precautions	: Do not let product enter drains.
Methods and materials for containment and cleaning up	: Cover drains. Collect, bind, and pump off spills. Observe possible material restrictions (see sections 7 and 10). Take up dry. Dispose of properly. Clean up affected area. Avoid generation of dusts.

SECTION 7. HANDLING AND STORAGE

For precautions see section 2.2.

Further information on storage conditions	: Tightly closed. Dry.
Storage class	: 13, Non Combustible Solids
Recommended storage temperature	: Recommended storage temperature see product label.
Packaging material	: Suitable material: LDPE Bottle/Jar, Poly Drum

SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Ingredients with workplace control parameters

Contains no substances with occupational exposure limit values.

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Engineering measures : No data available

Personal protective equipment

Respiratory protection : required when dusts are generated.

Our recommendations on filtering respiratory protection are based on the following standards: DIN EN 143, DIN 14387 and other accompanying standards relating to the used respiratory protection system.

Recommended Filter type: : Filter type P1

The entrepreneur has to ensure that maintenance, cleaning and testing of respiratory protective devices are carried out according to the instructions of the producer. These measures have to be properly documented.

Hand protection

Material : Nitrile rubber
Break through time : 480 min
Glove thickness : 0.11 mm
Protective index : Full contact
Manufacturer : KCL 741 Dermatril® L

Material : Nitrile rubber
Break through time : 480 min
Glove thickness : 0.11 mm
Protective index : Splash contact
Manufacturer : KCL 741 Dermatril® L

Remarks : This recommendation applies only to the product stated in the safety data sheet, supplied by us and for the designated use. When dissolving in or mixing with other substances and under conditions deviating from those stated in EN 16523-1 please contact the supplier of CE-approved gloves (e.g. KCL GmbH, D-36124 Eichenzell, Internet: www.kcl.de).

Eye protection : Use equipment for eye protection tested and approved under appropriate government standards such as NIOSH (US) or EN 166(EU).
Safety glasses

Hygiene measures : Change contaminated clothing. Wash hands after working with substance.

SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance	: Crystalline solid
Color	: colourless
Odor	: odourless
Odor Threshold	: Not applicable
pH	: 7
Melting point/ range	: 801 °C
Boiling point/boiling range	: 1,413 °C
Flash point	: Not applicable
Evaporation rate	: No data available
Flammability (solid, gas)	: The product is not flammable.
Flammability (liquids)	: No data available
Burning rate	: No data available
Upper explosion limit / Upper flammability limit	: No data available
Lower explosion limit / Lower flammability limit	: No data available
Vapor pressure	: No data available
Relative vapour density	: No data available
Relative density	: 2.16 (25 °C)
Density	: 2.16 g/cm ³ (25 °C)
Solubility(ies)	
Water solubility	: 317 g/l completely soluble (20 °C) pH: 7 - 10
Partition coefficient: n- octanol/water	: Not applicable for inorganic substances

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Autoignition temperature	: Not applicable
Decomposition temperature	: 1,000 °C
Viscosity, dynamic	: No data available
Viscosity, kinematic	: No data available
Flow time	: No data available
Explosive properties	: Not classified as explosive.
Oxidizing properties	: none
Surface tension	: 73.03 mN/m, 14.5 g/l, 23 °C
Molecular weight	: 58.44 g/mol
Particle characteristics	
Particle size	: No data available

SECTION 10. STABILITY AND REACTIVITY

Reactivity	: No data available
Chemical stability	: The product is chemically stable under standard ambient conditions (room temperature) .
Possibility of hazardous reactions	: Risk of explosion/exothermic reaction with: Alkali metals Exothermic reaction with: Lithium
Conditions to avoid	: no information available
Incompatible materials	: No data available
Hazardous decomposition products	: In the event of fire: see section 5

SECTION 11. TOXICOLOGICAL INFORMATION

11.1 Information on toxicological effects

Acute toxicity

Acute toxicity estimate Oral - > 2,000 mg/kg

(Calculation method)

LD50 Oral - Rat - male - 3,550 mg/kg

Remarks: (ECHA)

Inhalation: No data available

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LD50 Dermal - Rabbit - > 10,000 mg/kg

Remarks: (RTECS)

No data available

Skin corrosion/irritation

Skin - Rabbit

Result: No skin irritation

Remarks: (ECHA)

Serious eye damage/eye irritation

Eyes - Rabbit

Result: No eye irritation

Remarks: (ECHA)

Respiratory or skin sensitization

No data available

Germ cell mutagenicity

Animal testing did not show any mutagenic effects.

Test Type: Ames test

Test system: Escherichia coli/Salmonella typhimurium

Metabolic activation: with and without metabolic activation

Method: OECD Test Guideline 471

Result: negative

Carcinogenicity

Did not show carcinogenic effects in animal experiments.

Reproductive toxicity

Did not show teratogenic effects in animal experiments.

No impairment of reproductive performance suspected.

Specific target organ toxicity - single exposure

No data available

Specific target organ toxicity - repeated exposure

No data available

Aspiration hazard

No data available

11.2 Additional Information

RTECS: VZ4725000

Vomiting, Diarrhoea, Dehydration and congestion may occur in internal organs. Hypertonic salt solutions can produce inflammatory reactions in the gastrointestinal tract., Nausea

To the best of our knowledge, the chemical, physical, and toxicological properties have not been thoroughly investigated.

SECTION 12. ECOLOGICAL INFORMATION

Ecotoxicity

No data available

Persistence and degradability

No data available

Bioaccumulative potential

No data available

Mobility in soil

No data available

Other adverse effects

No data available

SECTION 13. DISPOSAL CONSIDERATIONS**Disposal methods**

Waste from residues : Waste material must be disposed of in accordance with the national and local regulations. Leave chemicals in original containers. No mixing with other waste. Handle uncleaned containers like the product itself.

SECTION 14. TRANSPORT INFORMATION**International Regulations****IATA-DGR**

Not regulated as a dangerous good

IMDG-Code

Not regulated as a dangerous good

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable for product as supplied.

National Regulations**TDG**

Not regulated as a dangerous good

Special precautions for user

Remarks : Not classified as dangerous in the meaning of transport regulations.

SECTION 15. REGULATORY INFORMATION**Canadian lists**

No substances are subject to a Significant New Activity Notification.

No substances are subject to CEPA Section 84 Ministerial Conditions.

SECTION 16. OTHER INFORMATION

Full text of other abbreviations

AIIC - Australian Inventory of Industrial Chemicals; ANTT - National Agency for Transport by Land of Brazil; ASTM - American Society for the Testing of Materials; bw - Body weight; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; ERG - Emergency Response Guide; GHS - Globally Harmonised System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organisation; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardisation; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; MERCOSUR - The Agreement for the Facilitation of the Transport of Dangerous Goods; n.o.s. - Not Otherwise Specified; Nch - Chilean Norm; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NOM - Official Mexican Norm; NTP - National Toxicology Program; NZIoC - New Zealand Inventory of Chemicals; OECD - Organisation for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; TCSI - Taiwan Chemical Substance Inventory; TDG - Transportation of Dangerous Goods; TECI - Thailand Existing Chemicals Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative; WHMIS - Workplace Hazardous Materials Information System

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